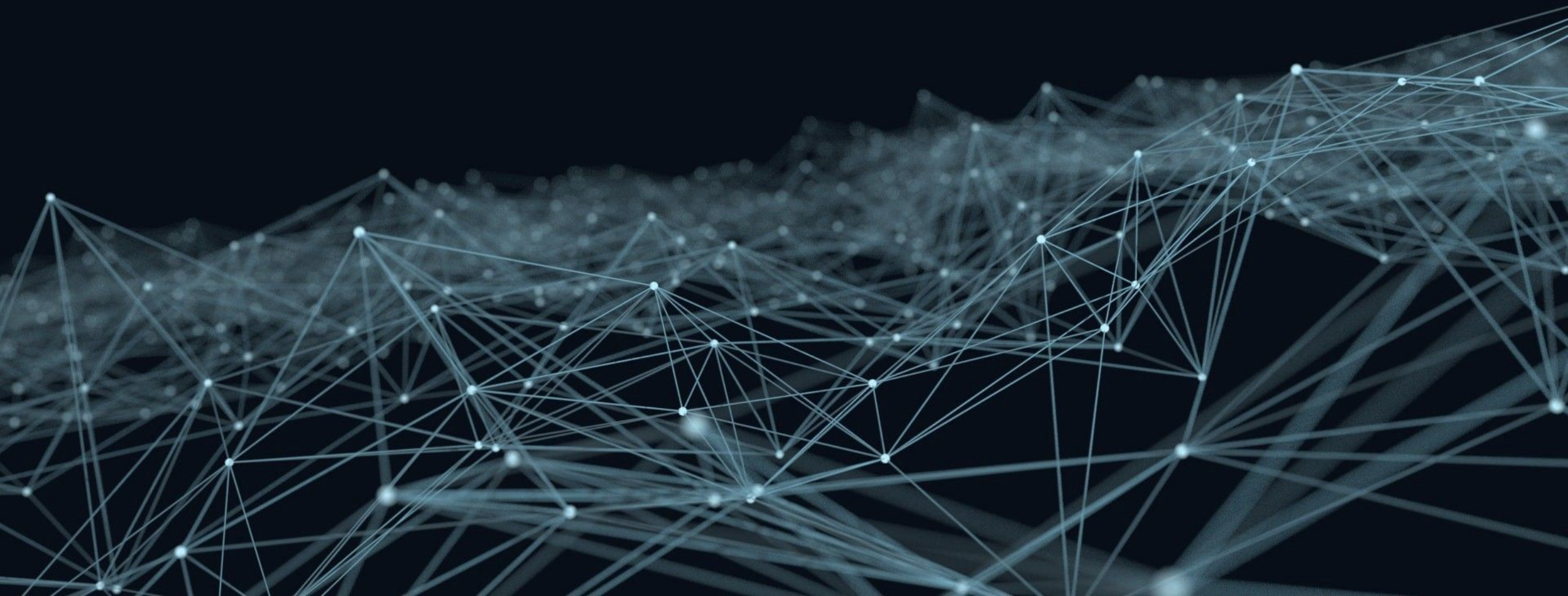
Java Full Stack Development

3. Working with Copilot



Module Topics

- Copilot IntelliJ Integration
- Understanding Copilot
- How Prompts Work
- Using Copilot to Write and Validate Code
- Prompt Engineering

Copilot IntelliJ Integration

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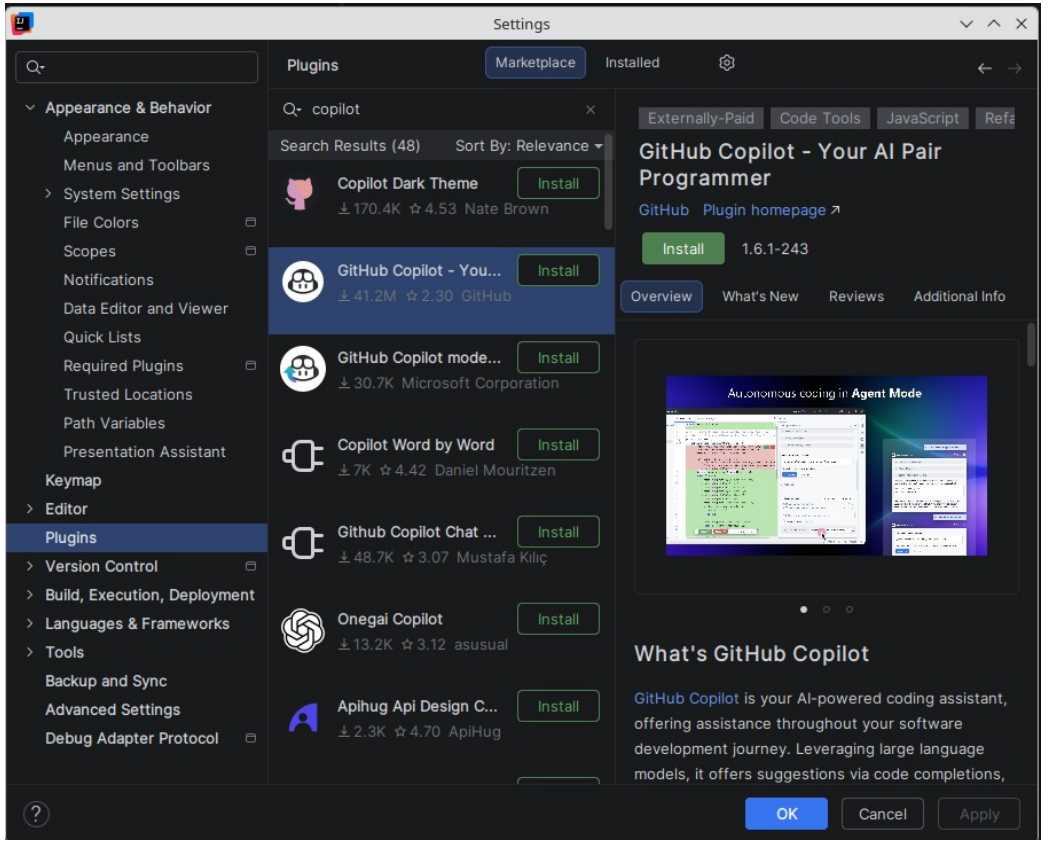
Copilot Integration

- GitHub Copilot integrates into IntelliJ
  - Via first-party JetBrains plugin
  - Developed and maintained by GitHub
- Available for IntelliJ IDEA
  - Both Community and Ultimate editions



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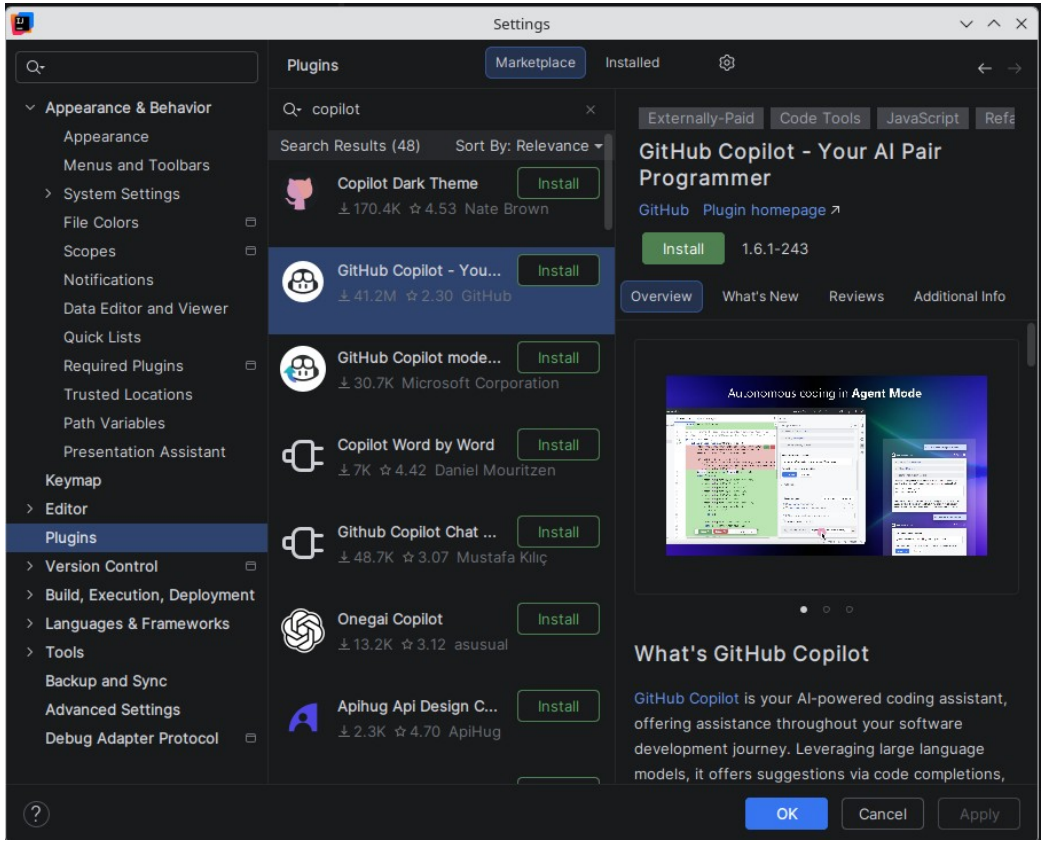
Copilot Integration

- Open up Plugins in settings
  - Search for GitHub Copilot
  - Install the plugin published by GitHub
  - Restart IntelliJ to activate
- Post-install
  - Sign in to GitHub
    - Tools → GitHub Copilot → Login to GitHub
    - A browser window opens for OAuth authentication
    - The IDE receives an access token on successful login



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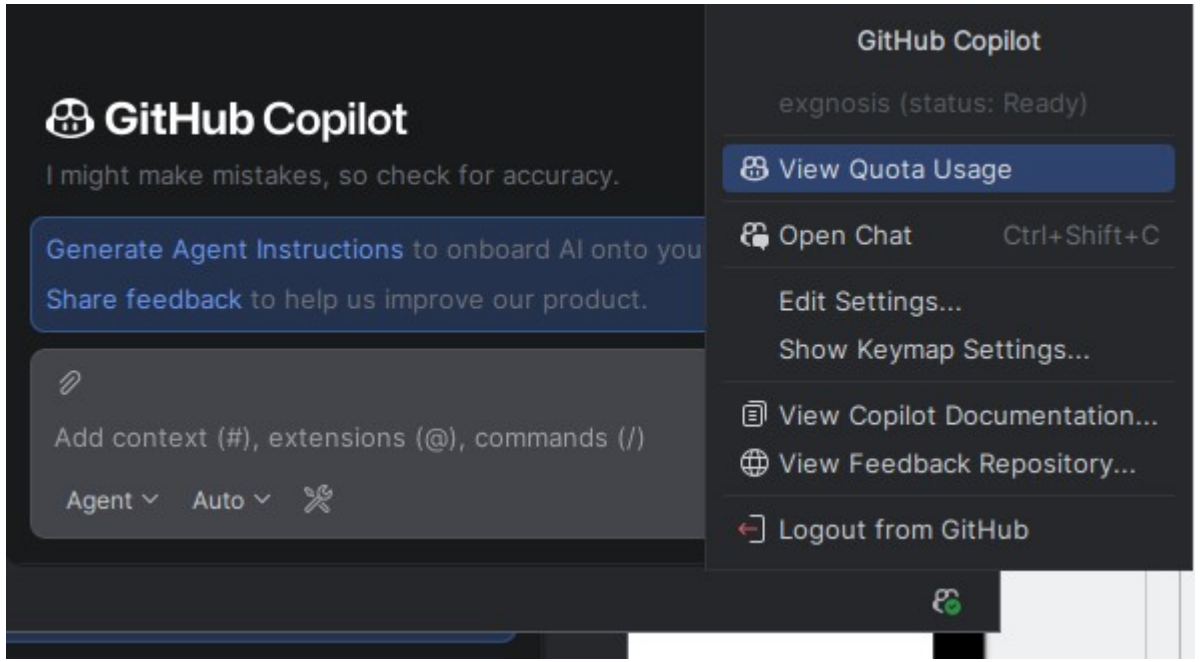
Copilot Integration

- Verify installation
  - Copilot icon appears in the status bar
    - Bottom right of the IDE
  - Status bar shows active/inactive state



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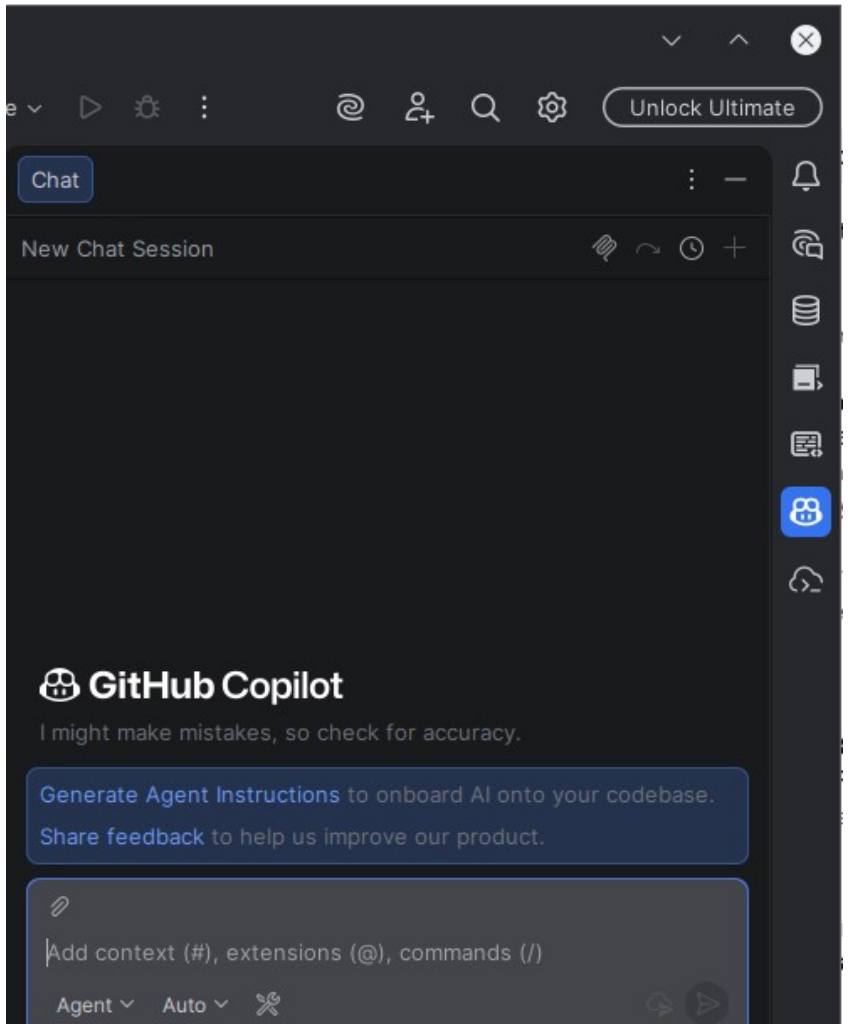
Copilot Integration

- Once installed, Copilot surfaces in two primary modes
  - Inline completions
    - Suggestions appear directly in the editor as you type
  - Copilot chat
    - A conversational interface docked inside the IDE
  - Both modes are available without leaving the editor
- Requires a GitHub Copilot license
  - Individual, Business, or Enterprise plan
  - Authentication is via your GitHub account



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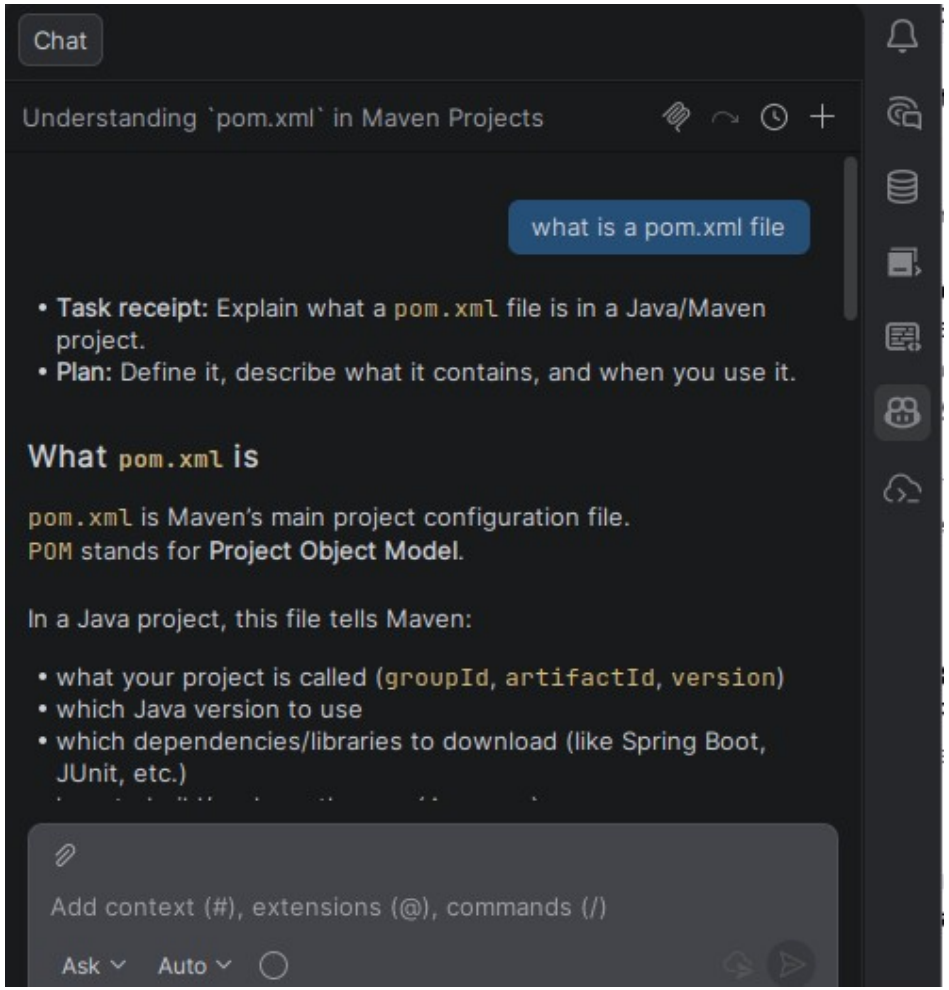
Copilot Integration

- Using chat mode is interactive
  - Shown in sidebar



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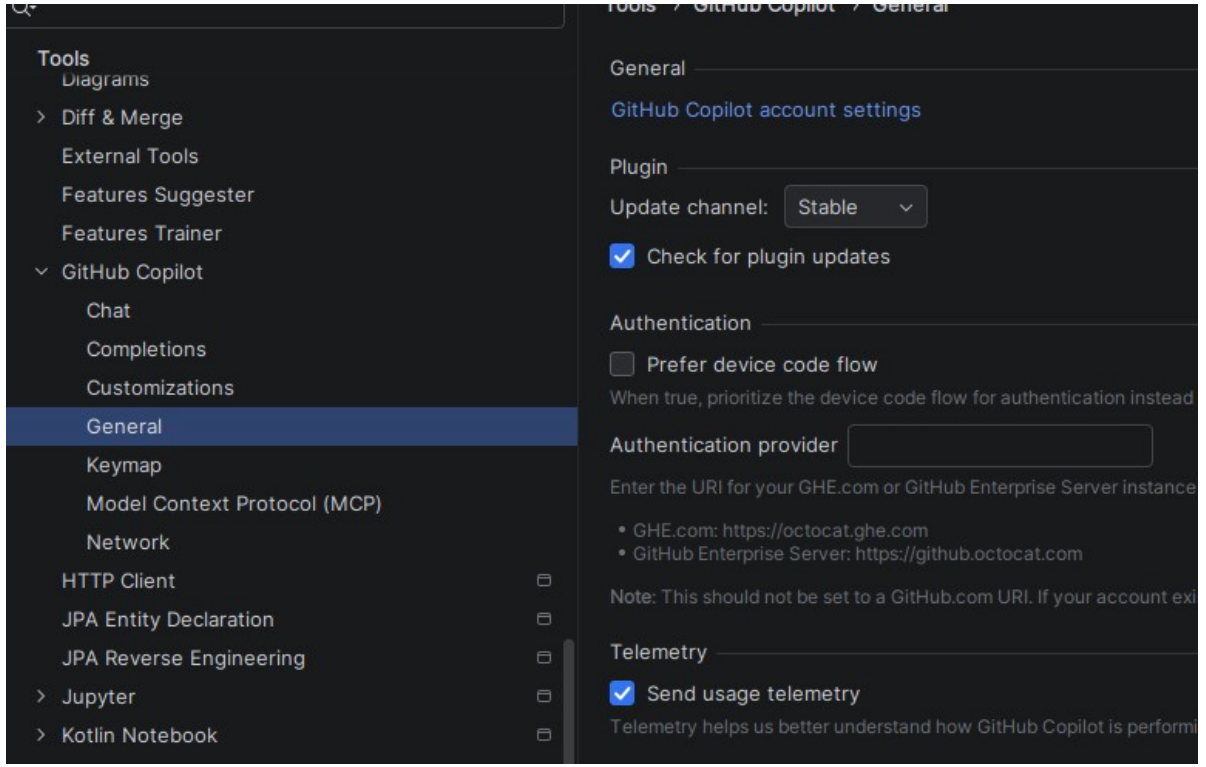
Copilot Integration

- Accessed via
  - Settings → Tools → GitHub Copilot
- Key settings
  - Enable/disable completions
    - Toggle inline suggestions globally or per language
  - Show completions automatically
    - Whether suggestions appear as you type or on demand
  - Duplication detection
    - Filter suggestions that closely match public code (if enabled by org)



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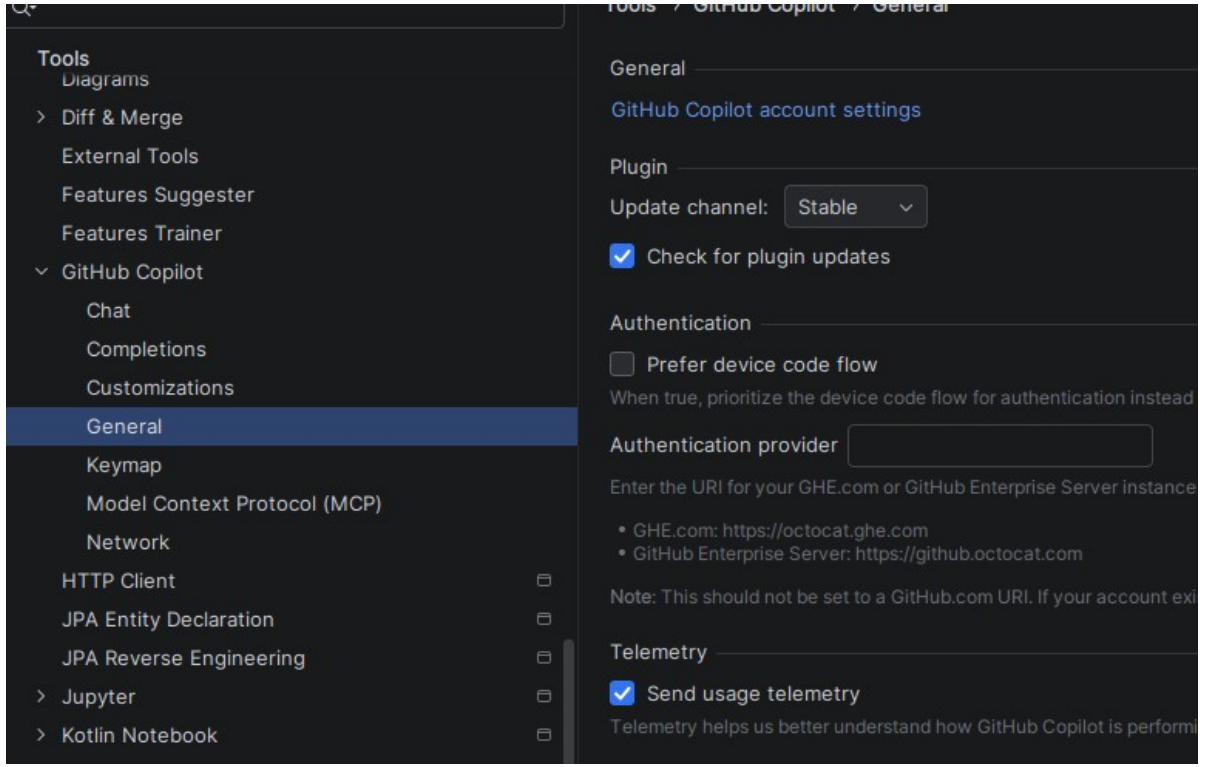
Copilot Integration

- Language-specific control
  - Completions can be enabled for Java but disabled for YAML or properties files
  - Useful if suggestions in configuration files create more noise than value
- Copilot Chat panel
  - Accessible via
    - View → Tool Windows → GitHub Copilot Chat
  - Can be docked to any side of the IDE



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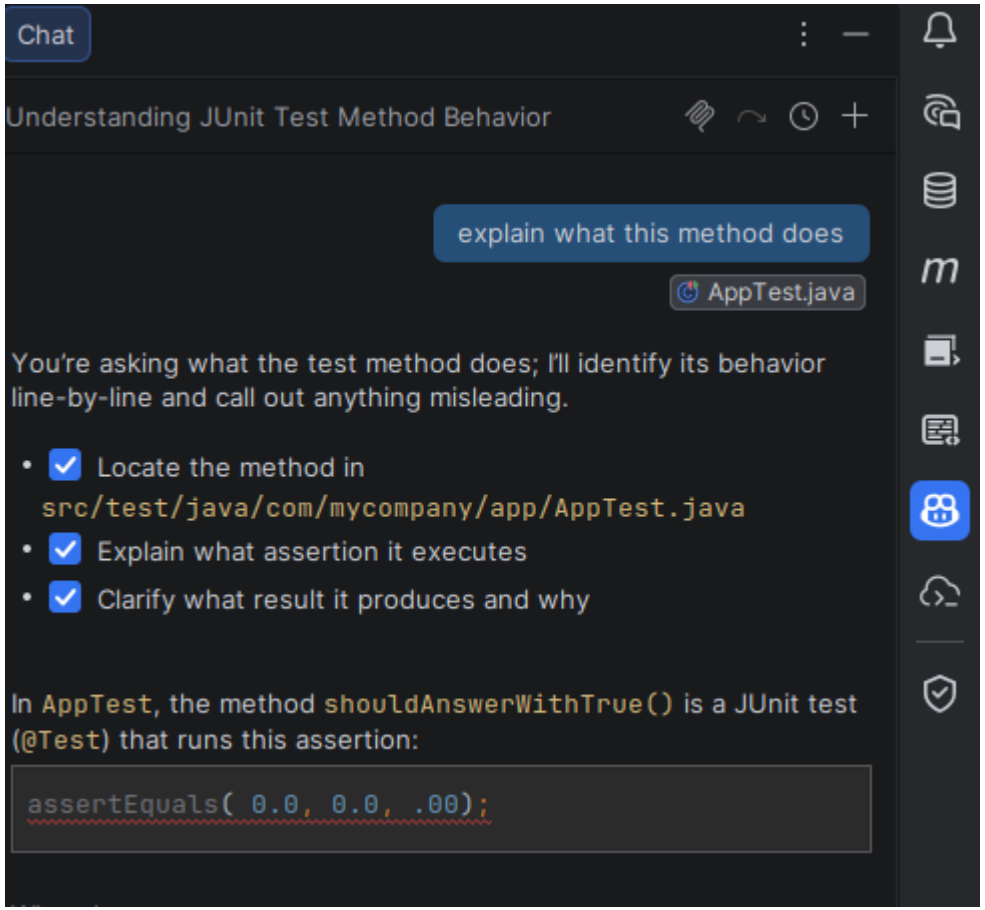
Copilot Chat Panel

- Conversational AI interface embedded in IntelliJ
  - Distinct from inline completions:
  - Completions suggest code as you type within the editor
  - Chat accepts free-form questions and returns detailed responses
- Chat is aware of
  - The file currently open in the editor
  - Selected code in the editor when you explicitly reference it
  - The project structure to a limited extent



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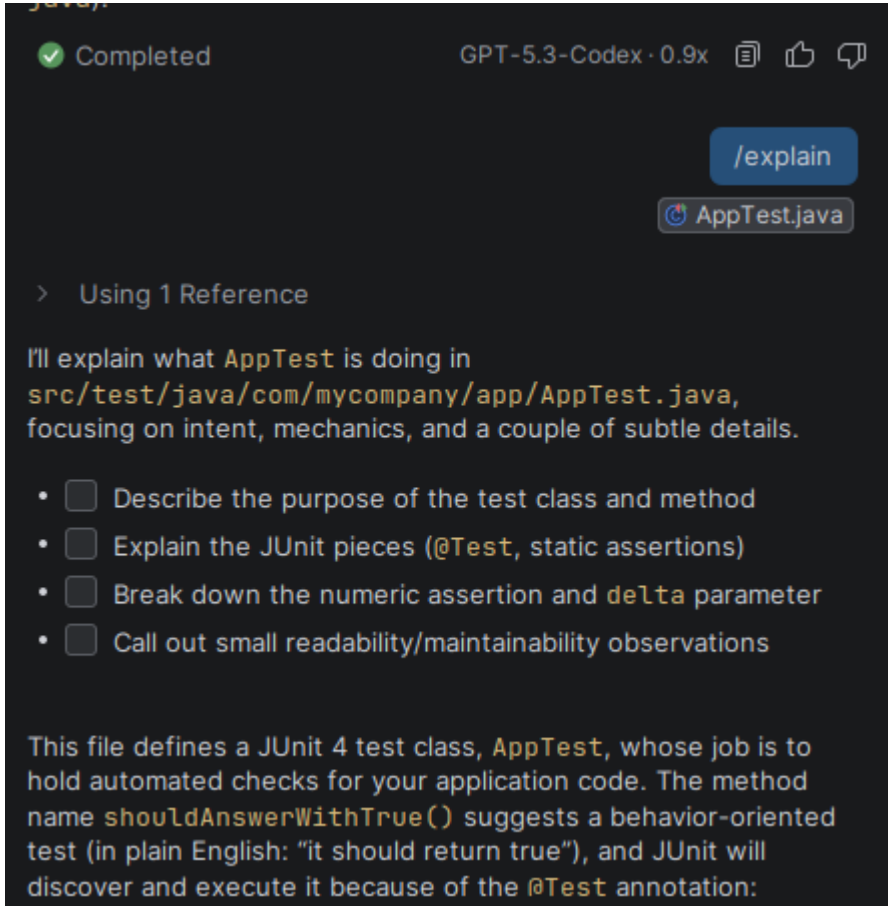
Copilot Chat Panel

- Slash commands in the Chat panel:
  - /explain
    - Explain selected code
  - /fix
    - Suggest a fix for a problem in selected code
  - /tests
    - Generate test cases for selected code
  - /doc
    - Generate Javadoc for selected code



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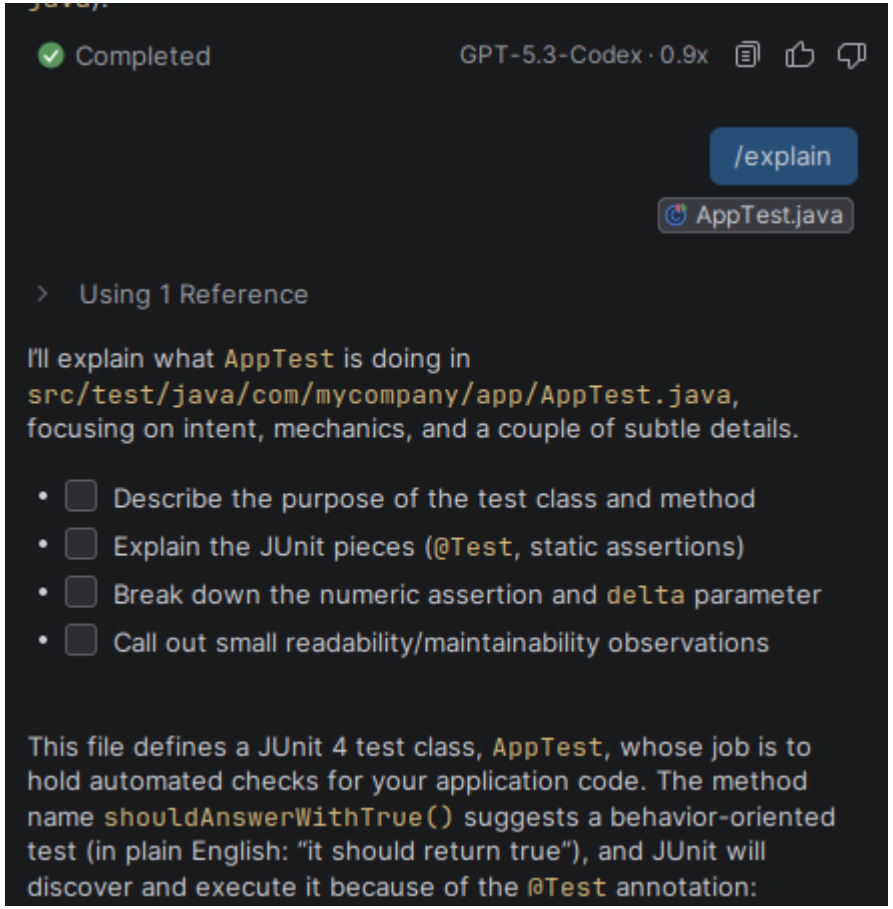
Copilot Chat Panel

- Completions are passive
  - They appear as you write and are accepted or dismissed with a keystroke
- Chat is active
  - You initiate a conversation with a specific goal Developers transitioning from text-editor
  - In practice, Chat is where the most significant productivity gains come from for complex tasks
    - Explaining unfamiliar Spring Boot patterns
    - Generating a full test class for an existing service
    - Asking for a refactored version of a method that is too complex



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Keyboard Shortcuts

- Common mistake
  - Using tab to accept a full suggestion
  - Crtl-Right for partial acceptance is often a more robust approach
    - Forces a more rigorous review of the suggestion
- Cycling through suggestions with Alt+] and Alt+[
  - Useful when the first suggestion is clearly wrong
  - Copilot typically has two to five alternatives for any given position

Action	Shortcut
Accept inline suggestion	`Tab`
Dismiss inline suggestion	`Escape`
Accept next word only	`Ctrl+Right` (Windows/Linux) / `Cmd+Right` (macOS)
Cycle to next suggestion	`Alt+]`
Cycle to previous suggestion	`Alt+[`
Trigger suggestion manually	`Alt+\`
Open Copilot Chat	`Ctrl+Shift+C` (configurable)
Explain selected code (Chat)	Select code → right-click → Copilot → Explain

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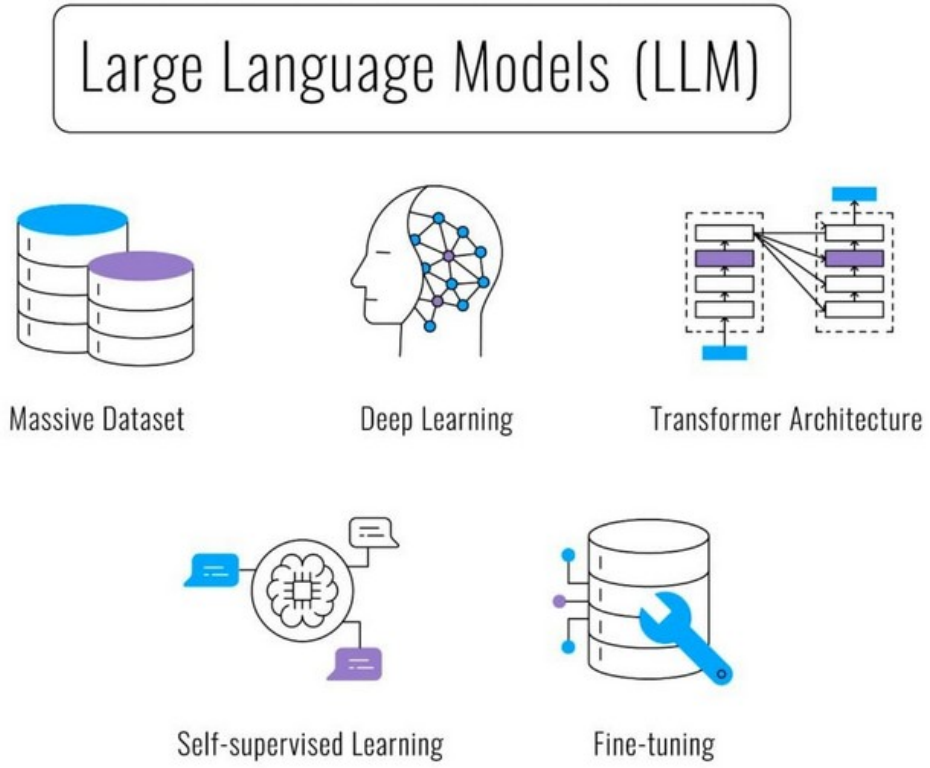
Understanding Copilot

GitHub Copilot

- AI-powered code completion tool
  - Built on a Large Language Model (LLM) trained primarily on source code
  - The underlying model is a descendant of the GPT model family
  - Trained on a large corpus of publicly available code from GitHub repositories, documentation, and technical text



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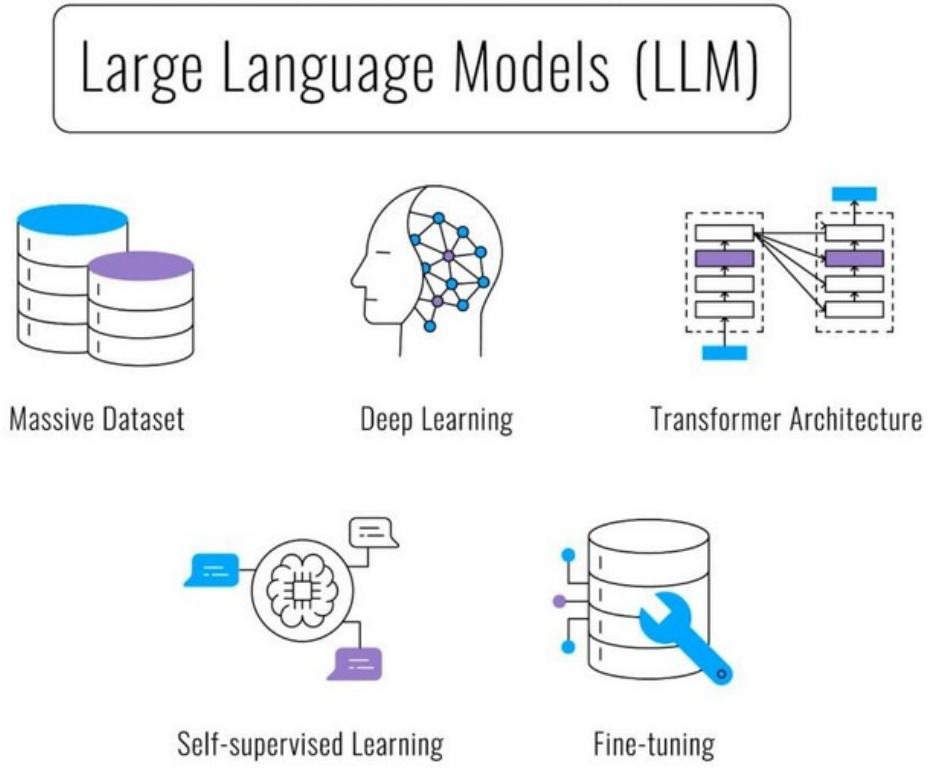
GitHub Copilot

- Copilot is a next-token prediction engine
  - Given a sequence of text consisting of your code, comments, and context
    - It predicts what text should come next
  - It does not understand your intent
    - It predicts what text is statistically likely to follow the text it has seen
    - It does not “understand” code
- It is not
  - A compiler or static analyzer
  - A search engine that retrieves existing code
  - A system that reasons about correctness



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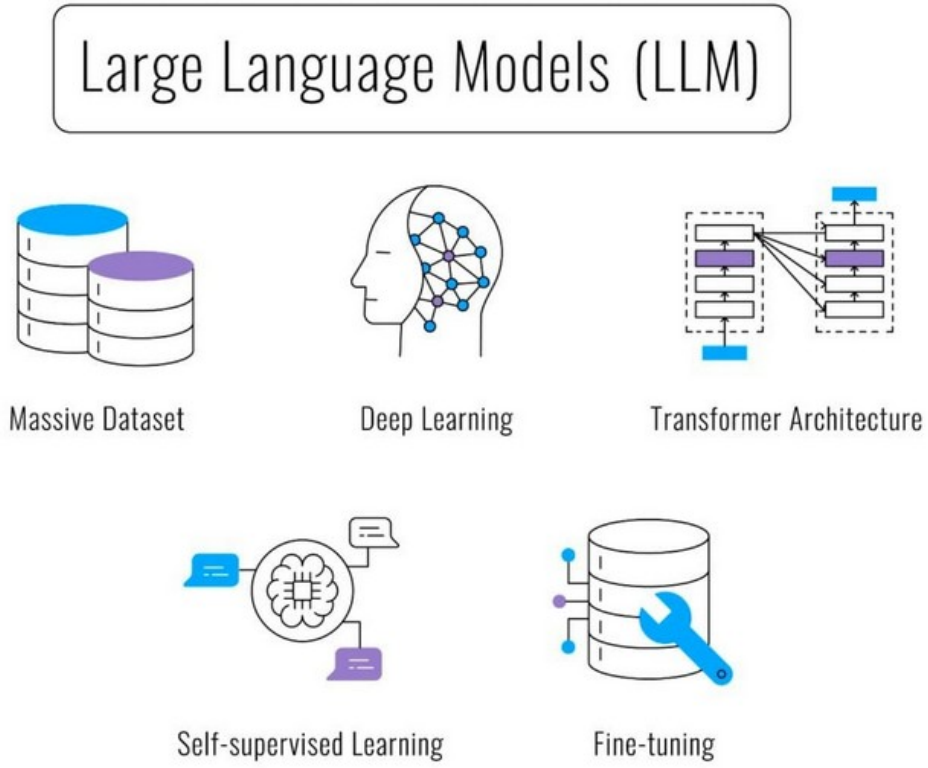
Training Data Foundation

- Copilot was trained on a large corpus of publicly available code
  - Billions of lines of code across many languages, with Java being heavily represented
  - Also includes Stack Overflow content, documentation, and technical articles
- Quality of suggestions correlates with
  - How common the pattern is in training data
  - How much relevant context exists in the current file and surrounding code



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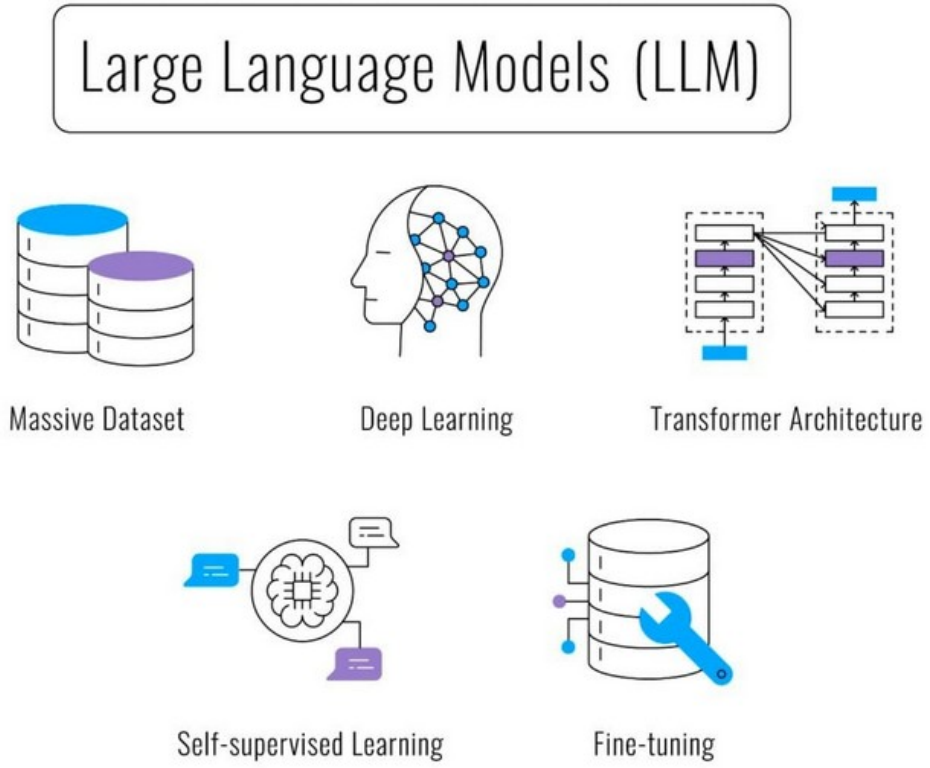
Training Data Foundation

- Implications of training data for Java and Spring Boot
  - Common patterns are well-represented
    - REST controllers, JPA repositories, service layers
  - Standard annotations are understood contextually
    - @RestController, @Service, @Transactional
  - Recent framework versions may be underrepresented
  - Organization-specific patterns and custom frameworks are not in the training data



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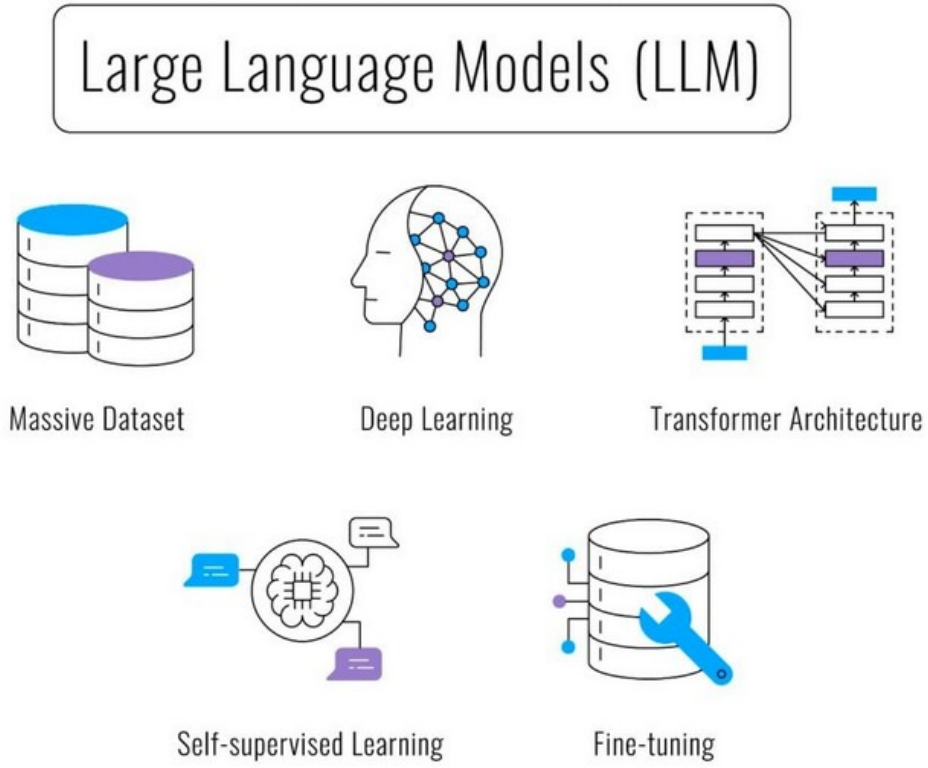
Training Data Foundation

- For Java and Spring Boot specifically
  - Copilot's training data coverage is strong for the standard patterns
    - CRUD operations
    - REST endpoint definitions
    - JPA query methods
    - Exception handling
    - Unit test structure
  - It performs less reliably on
    - Specialized domains
    - Custom security filter implementations
    - Complex Spring batch configurations
    - Oracle-specific SQL optimizations
    - Kafka consumer error handling with specific retry semantics
-



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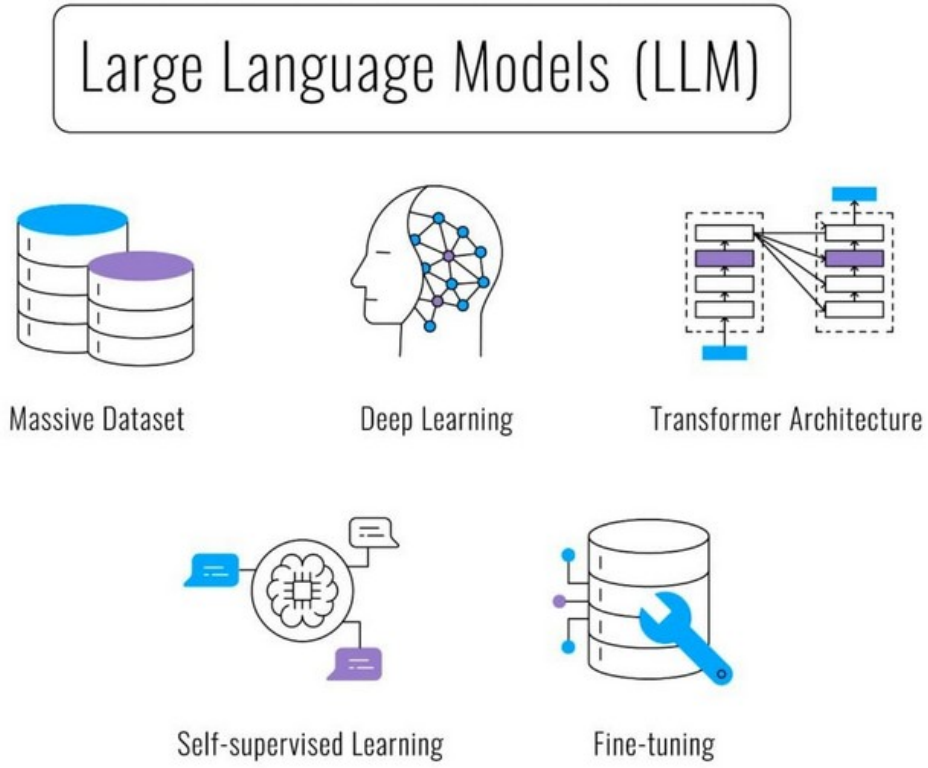
Training Data Foundation

- Where training coverage is weak
  - Copilot can still provide a useful starting point
  - But the developer's expert knowledge turn the generic suggestion into the correct implementation



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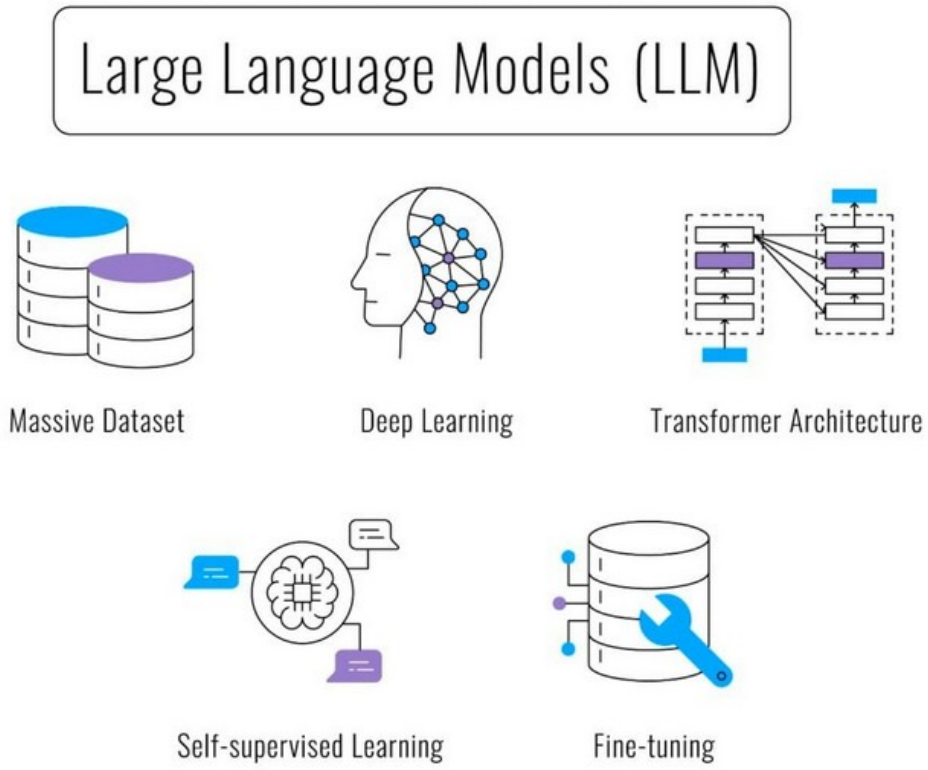
The Context Window

- Copilot's suggestions are determined by what it can see
  - Called the context window
  - Context includes
    - The current file
    - Everything above and below the cursor
    - The filename and path, used to infer intent
    - Recently opened files in the same IDE session
    - Selected code or text when using Chat



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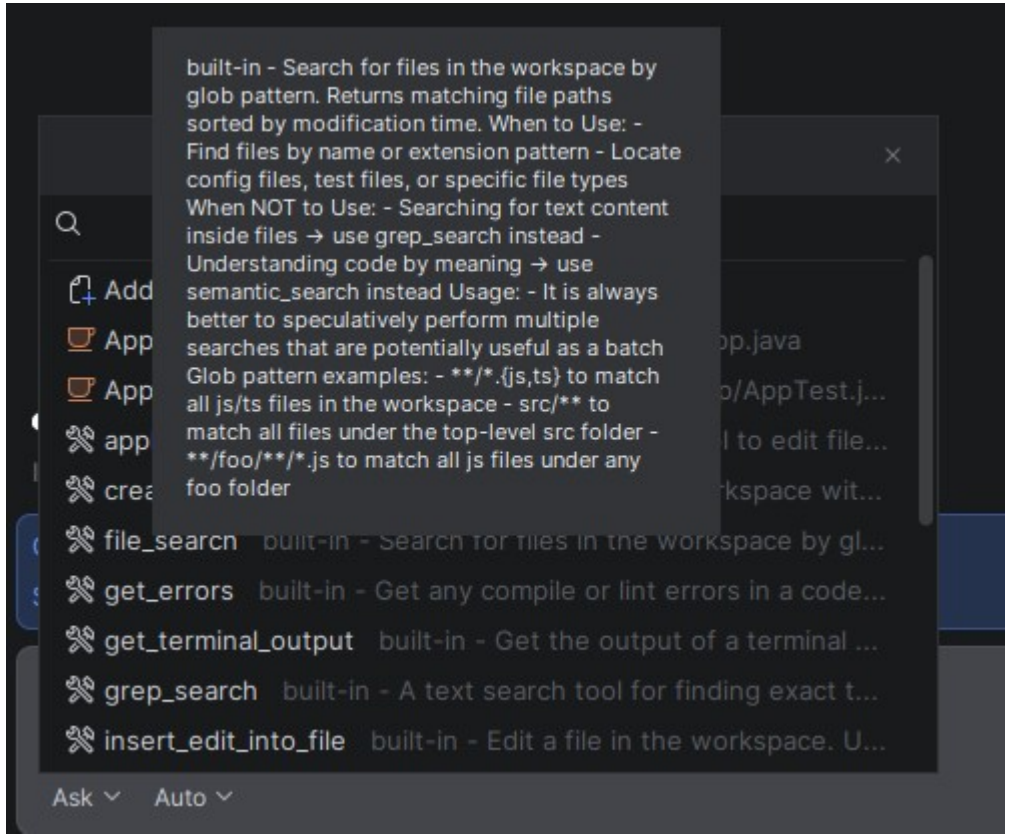
The Context Window

- Context
  - Does not automatically include
    - Other files in your project that you have not opened
    - Your application.yml configuration
    - Your database schema
    - Business rules that exist only in documentation or in someone's head
  - The quality of Copilot's output is directly constrained by the quality of its context
  - You can add artifacts to the context



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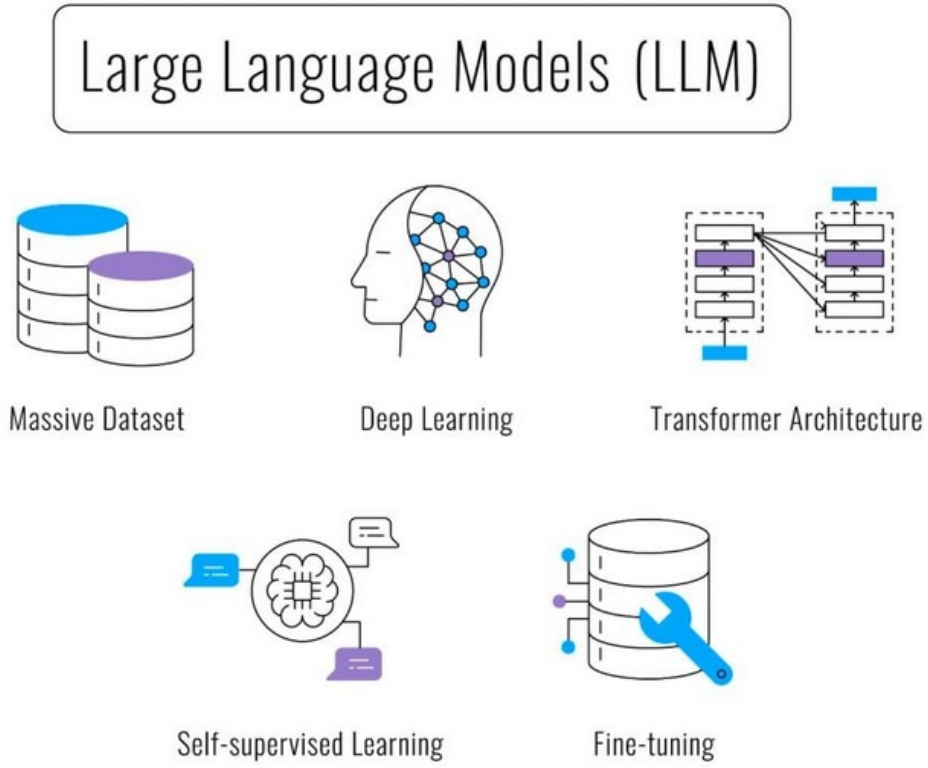
Capabilities

- Well-suited tasks
  - Generating boilerplate
    - getters, constructors, equals/hashCode, toString
  - Scaffolding standard patterns
    - REST controllers, service classes, JPA repositories
  - Writing unit tests for methods with clear input/output contracts
  - Completing well-established algorithm patterns
  - Generating Javadoc from method signatures and parameter names
  - Translating a descriptive comment into a code implementation



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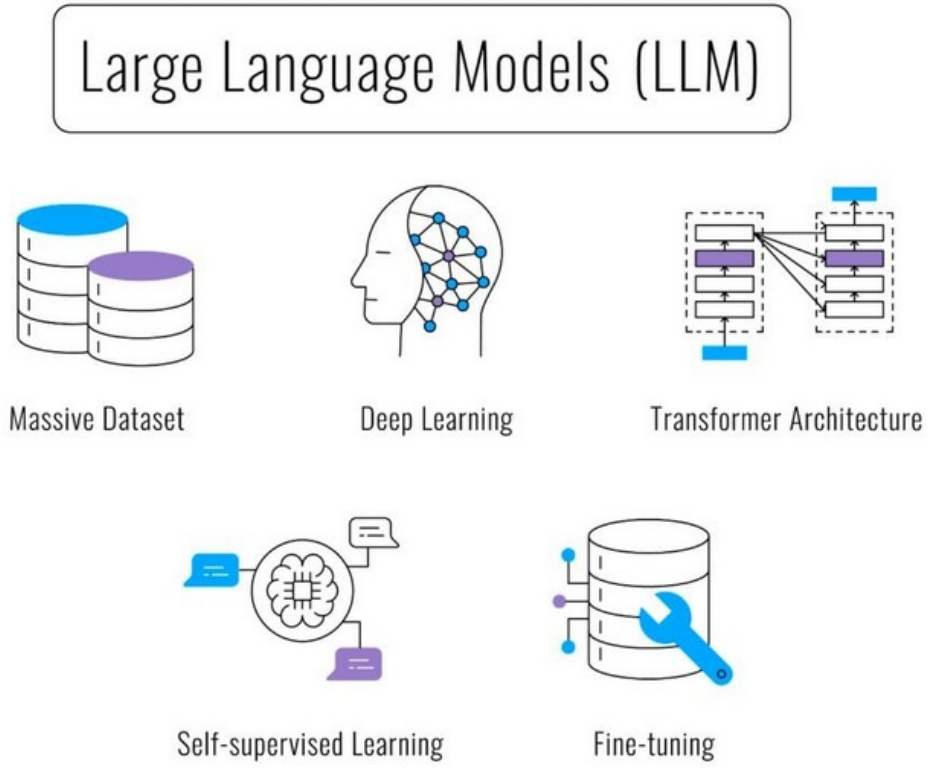
Capabilities

- Poorly-suited tasks
  - Reasoning about correctness in novel or domain-specific logic
  - Understanding cross-file dependencies it has not seen
  - Producing correct implementations of security-sensitive code without explicit guidance
  - Making architectural decisions
  - Replacing code review



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How Prompts Work

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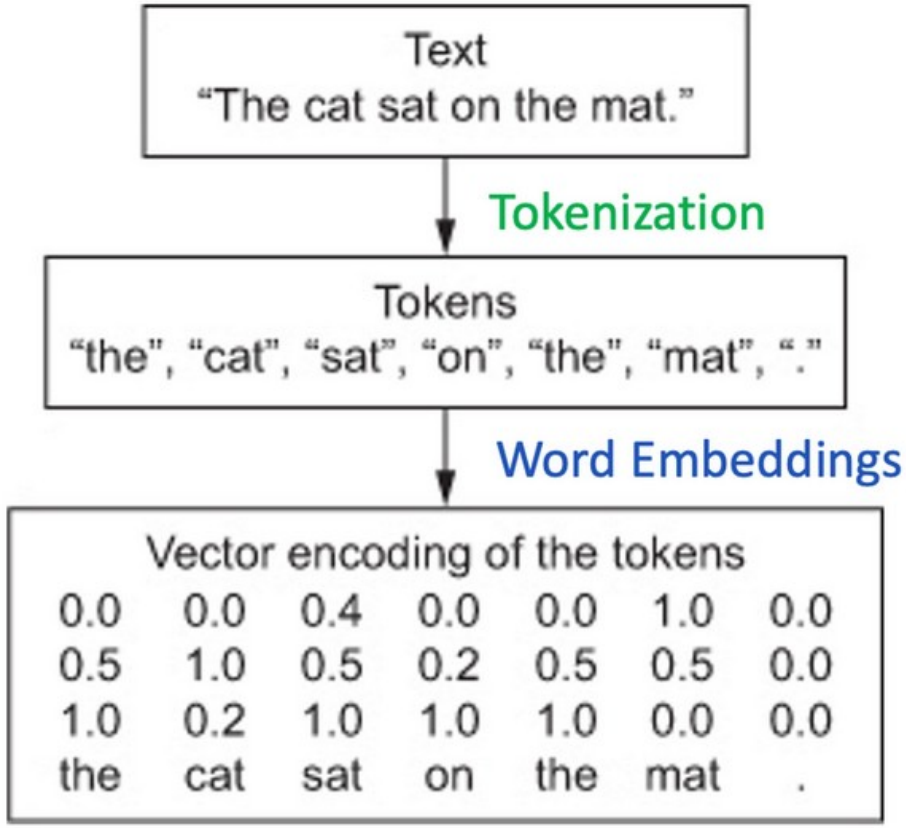
Tokens

- LLMs
  - Do not process characters, words, or lines
  - They process tokens
  - A token is roughly 3–4 characters of text on average
    - Common words are single tokens; rare words or identifiers may be multiple tokens
    - Code identifiers, keywords, punctuation, and whitespace are all tokenized



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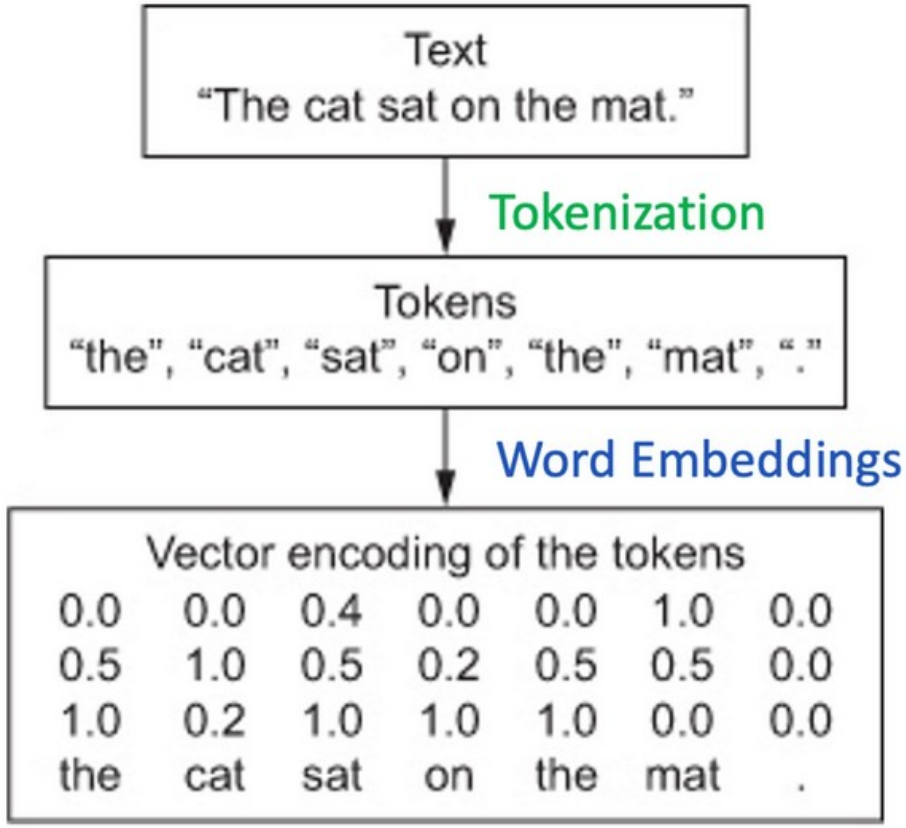
Tokens

- LLMs
  - Copilot's context window is measured in tokens, not characters or lines
    - A file with 500 lines of Java code may consume 4,000–8,000 tokens
    - Copilot can only "see" the tokens that fit within its context window
  - Practical implication:
    - In a very large file, code far from the cursor may fall outside the context window
    - Copilot weighs tokens closer to the cursor more heavily than those further away



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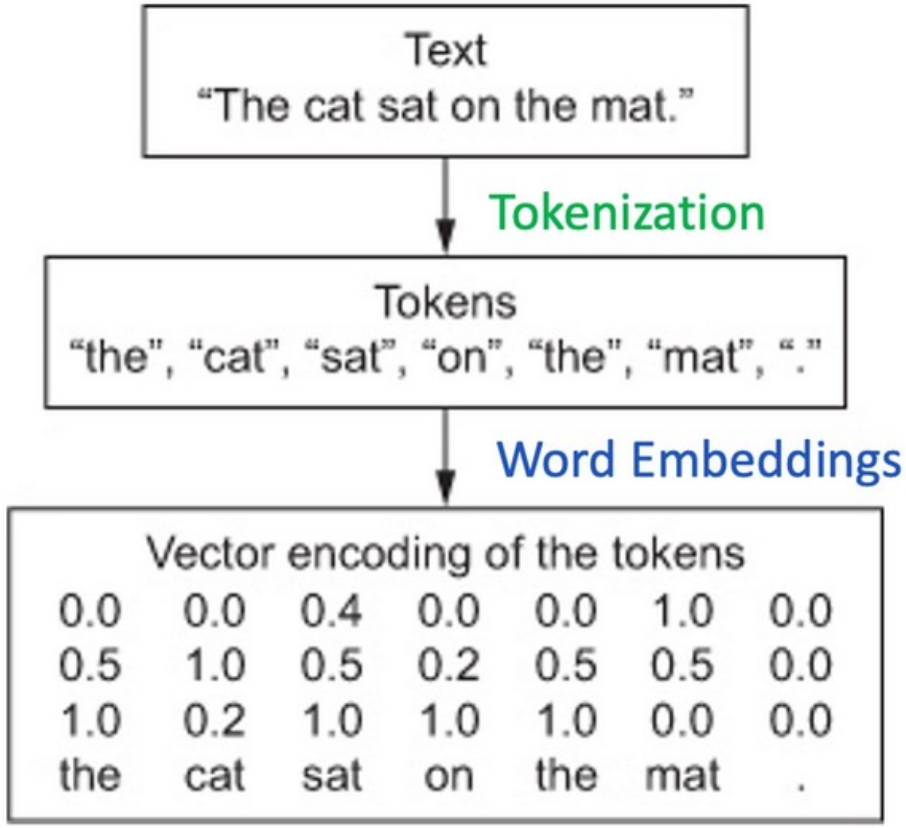
Tokens

- Implications
  - If a Java class is 800 lines long and the cursor is near the bottom
    - The class-level fields declared at the top may be partially or fully outside the context window by the time Copilot is generating a suggestion near the end of the file
  - This is one reason to prefer smaller, well-focused classes
  - A 400-line God class receives worse AI assistance than the same logic split into four 100-line focused classes
    - Each of which fits comfortably inside the context window



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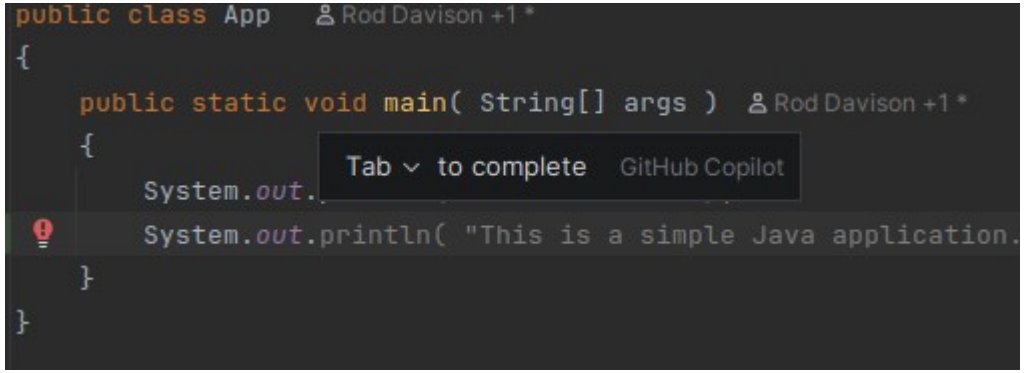
Completion

- When you pause typing
  - Copilot sends context to the model and receives a completion
- The process
  - IntelliJ plugin assembles the context
    - Current file, cursor position, recently opened files
  - Context is tokenized and sent to the Copilot model
    - The model is running on GitHub's servers
  - The model generates a probability distribution over possible next tokens
  - The most probable sequence of tokens is returned as the suggestion
  - The plugin renders this as greyed-out text in the editor



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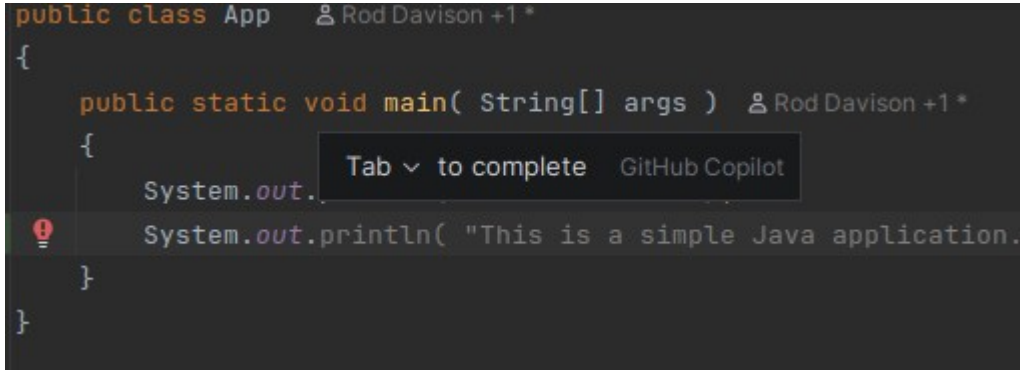
Completion

- The model does not run locally
  - Suggestions require an internet connection
  - Latency is typically under one second for inline completions



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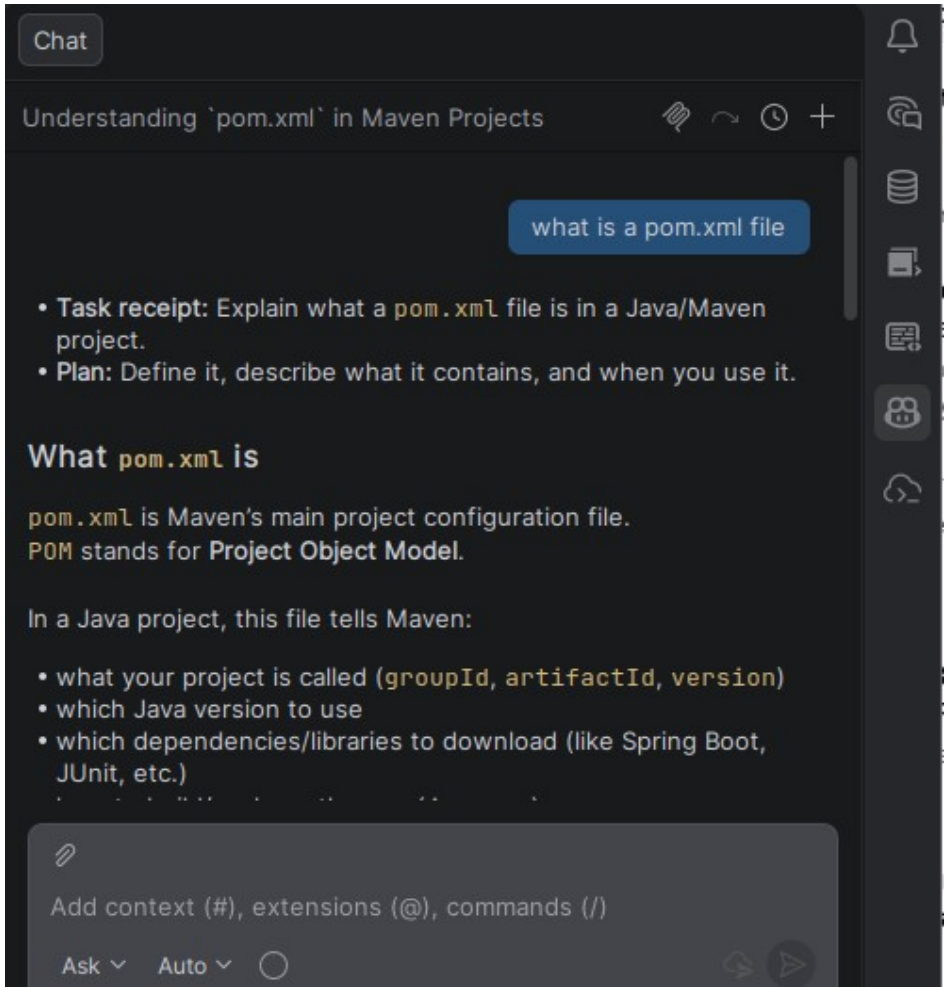
Chat Interactions

- Copilot Chat
  - Uses a conversational prompt-response model
  - You write a natural-language prompt
    - A question, instruction, or description
  - The model generates a response based on the prompt plus context
- The model sees
  - Your prompt text
  - The current file open in the editor
    - Automatically included
  - Any code you have explicitly selected or referenced with # syntax
  - The conversation history within the current Chat session



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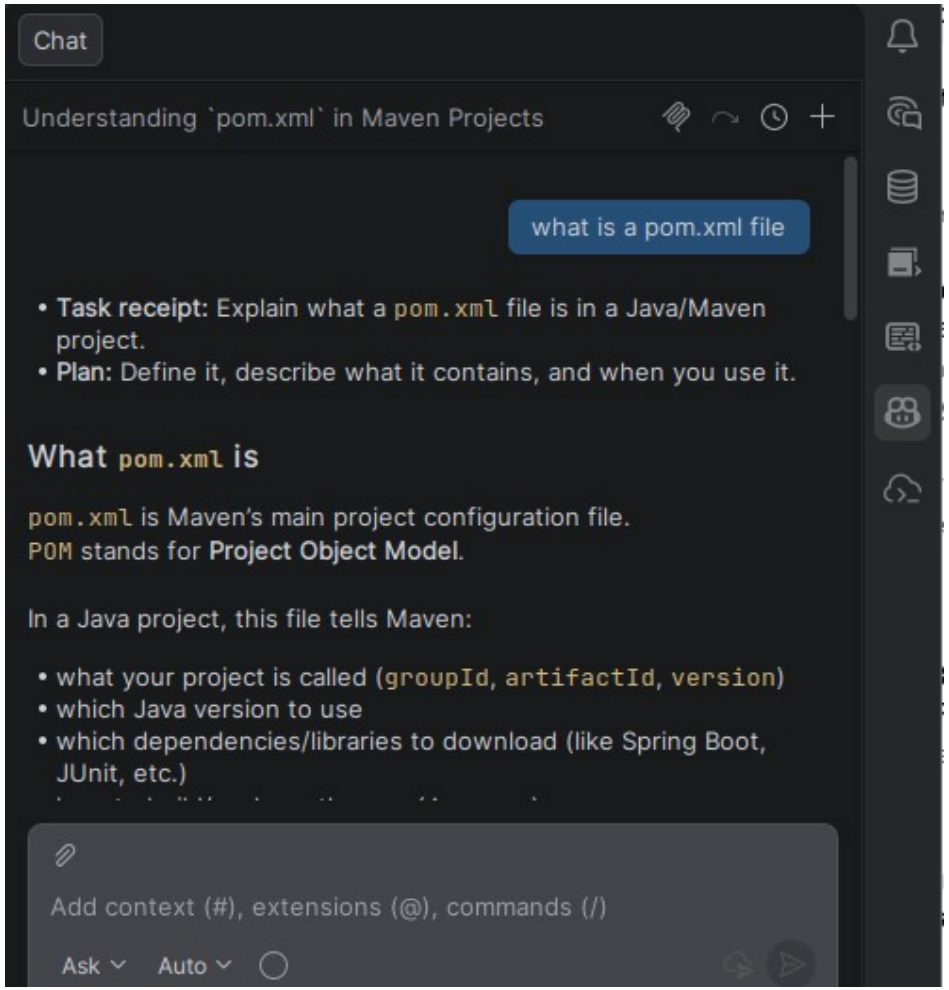
Chat Interactions

- Chat responses can include:
  - Natural-language explanations
  - Code blocks
    - Java, YAML, SQL, or any language
  - Mixed text and code



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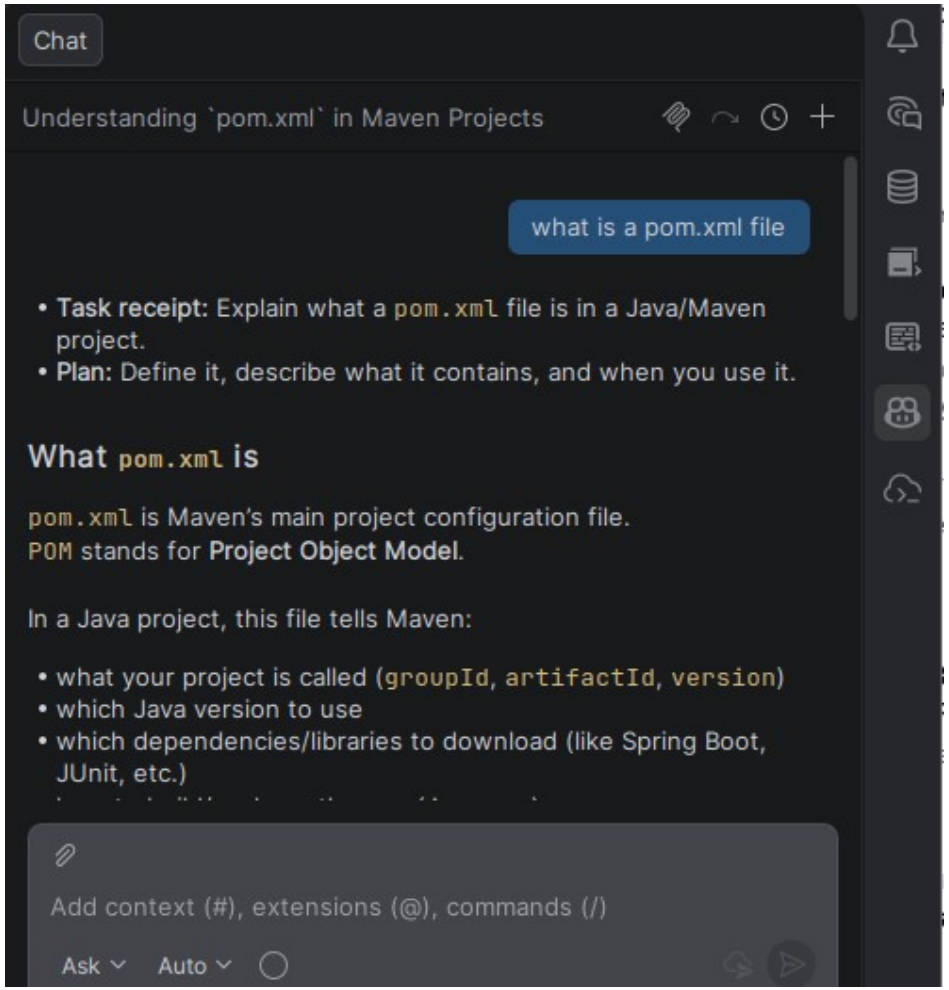
Chat Interactions

- Typing # in the Chat input
  - Opens a picker that lets you reference specific files, symbols, or even the entire project
  - For example
    - Typing "Refactor #OrderService to use the repository pattern"
    - Tells the model explicitly which file to focus on
    - Without the reference, the model uses whatever is in the active editor



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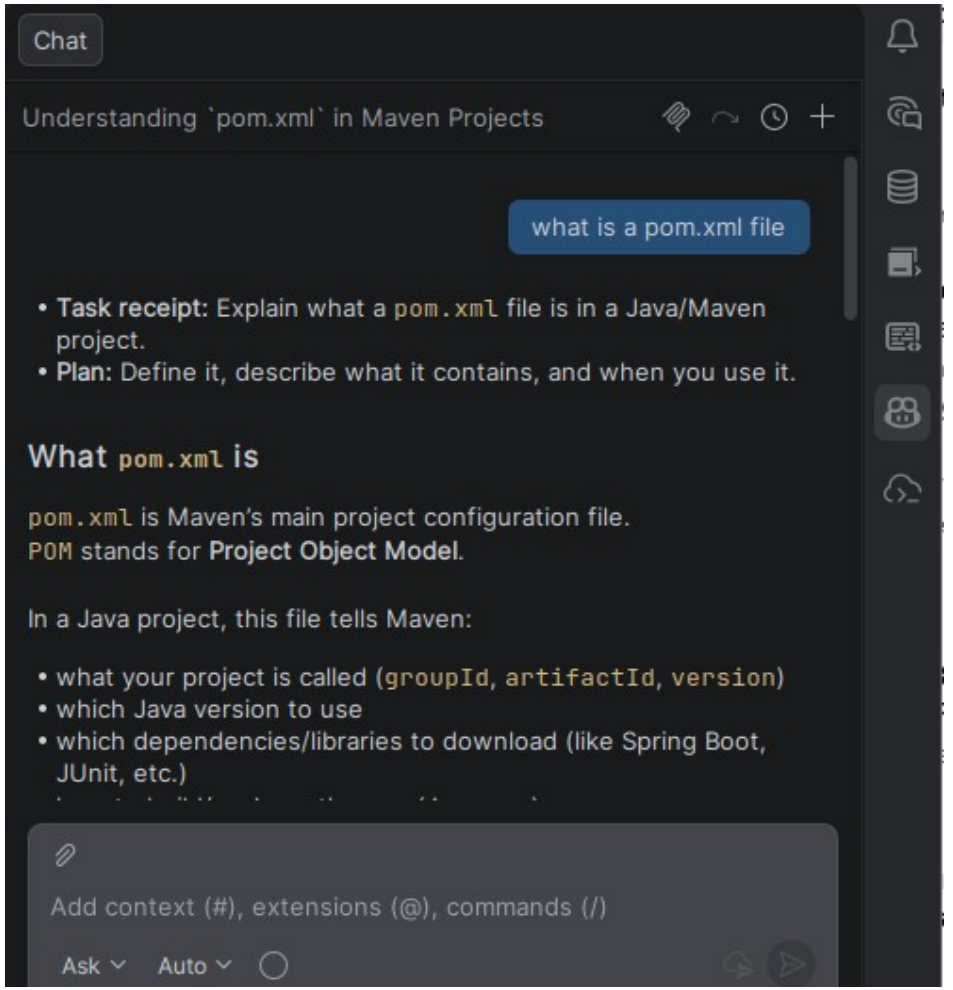
Same Prompt Different Outputs

- LLMs have a temperature parameter that controls output randomness
  - Higher temperature: more varied, creative outputs
  - Lower temperature: more deterministic, predictable outputs
  - Copilot is tuned for a moderate temperature appropriate for code generation



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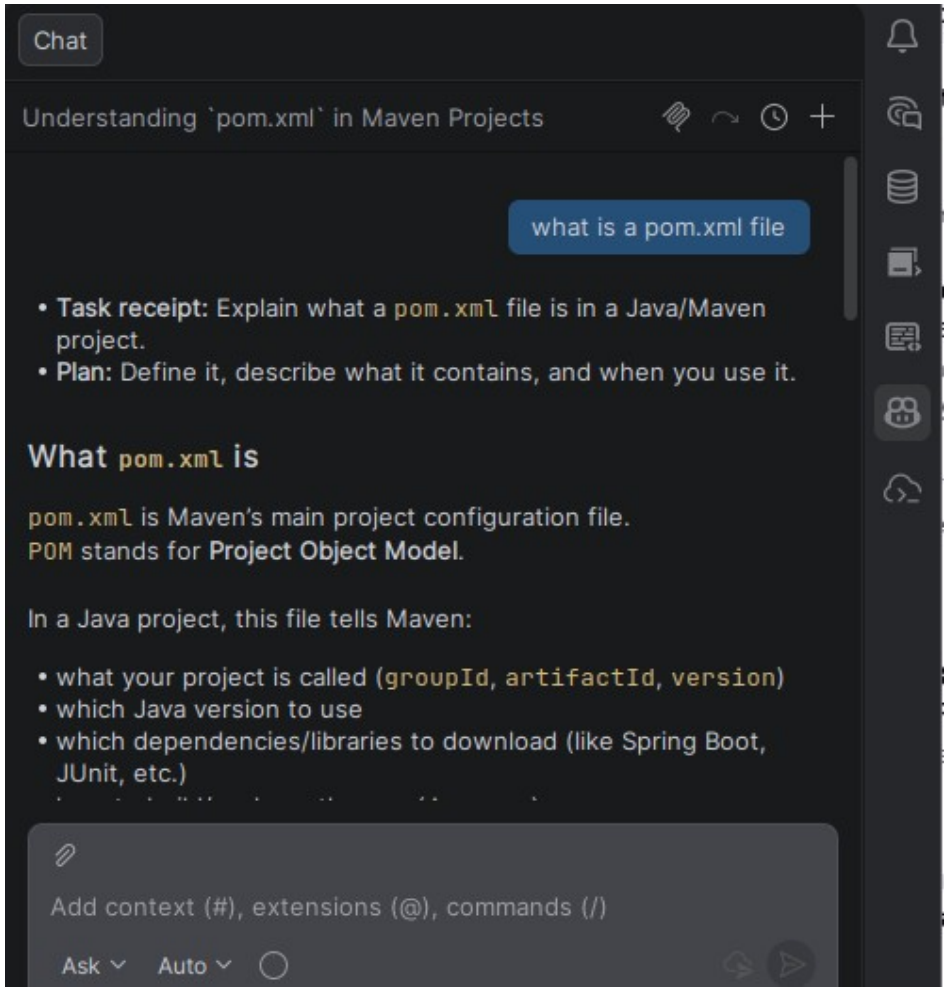
Same Prompt Different Outputs

- Practical consequence
  - Running the same prompt multiple times may produce different results
    - Use Alt+] / Alt+[to cycle through alternative suggestions
    - In Chat, you can ask again or ask Copilot to try a different approach
  - This is not a bug
    - It reflects the probabilistic nature of the underlying model
    - It means never assume the first suggestion is the only possible suggestion



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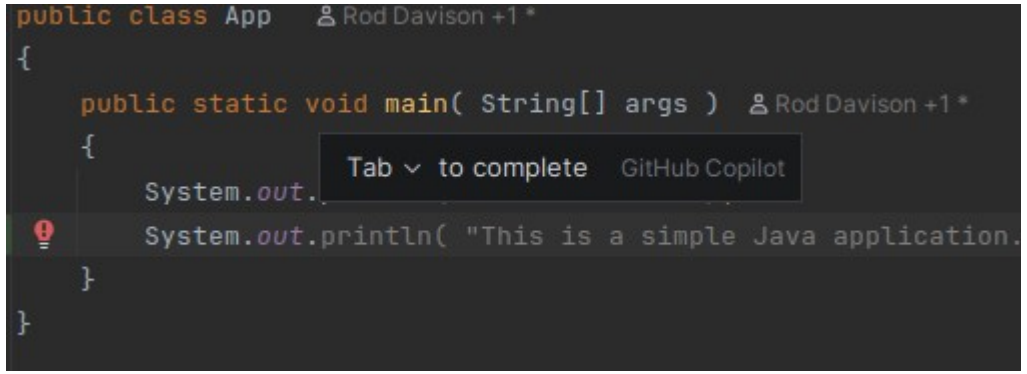
Using Copilot to Write and Validate Code

Inline Completions

- The primary Copilot interaction during active coding
- Workflow
  - Write a method signature or a descriptive comment
  - Pause while Copilot generates a suggestion as greyed-out text
  - Read the suggestion before accepting
  - Accept (Tab), accept word-by-word (Ctrl+Right), cycle alternatives (Alt+]), or dismiss (Esc)



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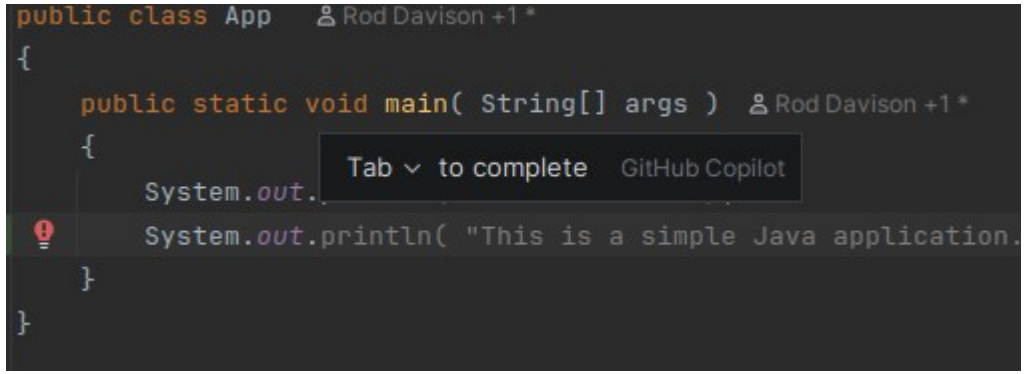
Inline Completions

- Copilot is most effective when given clear signals
  - A descriptive method name that conveys intent
  - A typed return type that constrains the output
  - A Javadoc comment describing the expected behaviour
  - An existing similar method in the same class (pattern matching)



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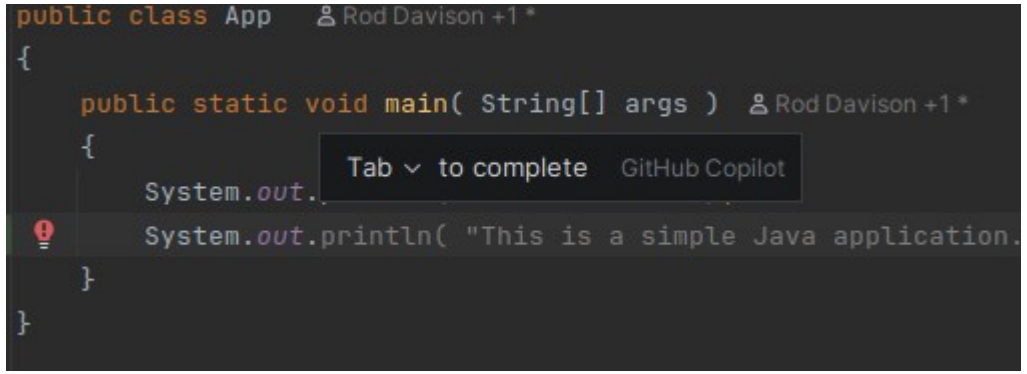
Inline Completions

- Caveat
  - Tab completion in traditional IDEs is safe to accept without reading
    - Because the IDE is drawing from the resolved type graph
    - It can only offer valid completions
  - Copilot suggestions are probabilistic and can be wrong.
    - Treat every Copilot suggestion as a code review of code written by a fast but fallible junior developer
    - Accept or reject based on your understanding, not on its plausibility



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Comment-Driven Code

- Writing a descriptive comment
  - Describes what the code does
  - The comment acts as a specification
    - The more precise it is
    - The more accurate the suggestion
  - Useful for
    - Complex query methods that would require significant thought to write from scratch
    - Algorithm implementations where the logic can be stated clearly in plain language
    - Utility methods with well-defined input/output contracts

```
// Find all orders placed in the last 30 days with a total value
// greater than the specified threshold. Return sorted by date descending.
public List<Order> findRecentHighValueOrders(BigDecimal threshold) {
    // Copilot generates the method body here
}
```

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Comment-Driven Code

- Inverts the traditional workflow
  - Instead of writing code and then documenting it
  - You write the documentation first and let Copilot write the code
- Secondary benefit beyond productivity
  - It forces you to articulate exactly what the method should do before writing it
  - Developers who struggle to write a clear comment describing a method's behaviour
    - Often discover that they have not yet fully thought through what the method should do

```
// Find all orders placed in the last 30 days with a total value
// greater than the specified threshold. Return sorted by date descending.
public List<Order> findRecentHighValueOrders(BigDecimal threshold) {
    // Copilot generates the method body here
}
```

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Comment-Driven Code

- The comment-first discipline
  - Lightweight version of test-driven development's *"specify before you implement"* principle
- In a Spring Data JPA context
  - A comment describing the query semantics often produces a correct @Query annotation or repository method name on the first suggestion

```
// Find all orders placed in the last 30 days with a total value
// greater than the specified threshold. Return sorted by date descending.
public List<Order> findRecentHighValueOrders(BigDecimal threshold) {
    // Copilot generates the method body here
}
```

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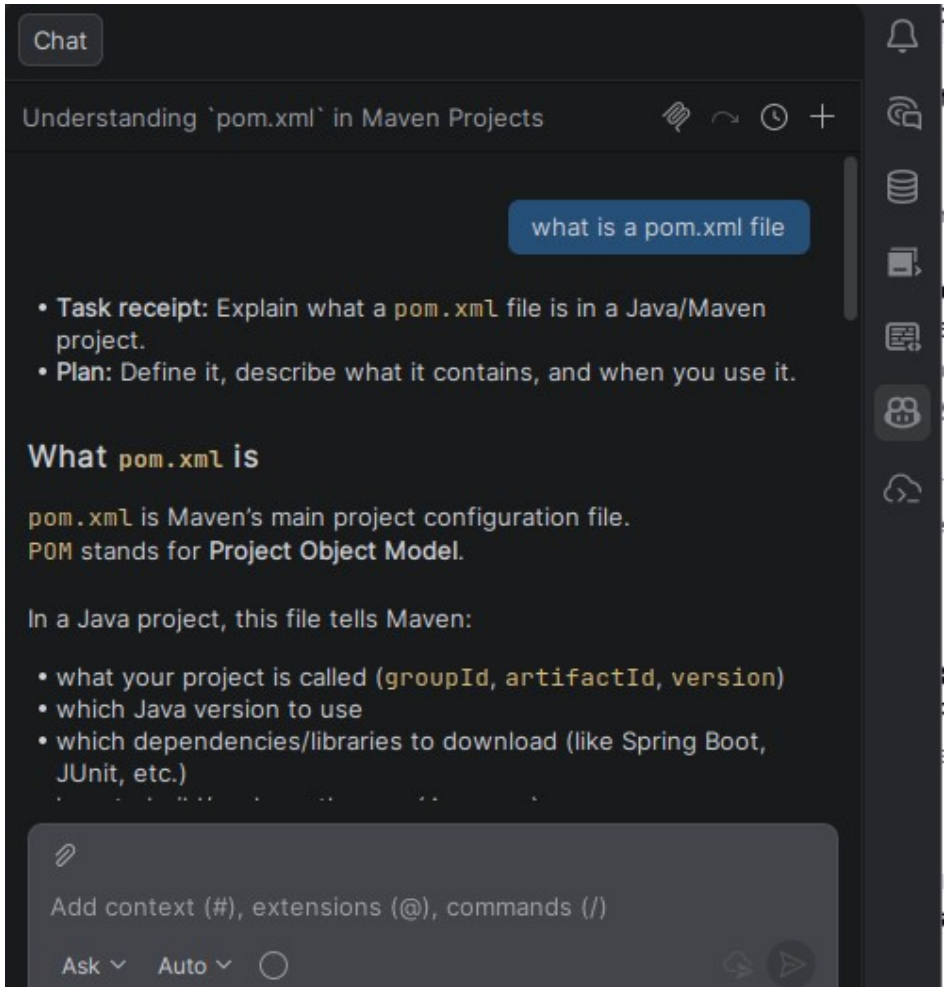
Writing Tests

- Approaches
  - Chat slash command
    - Select a method or class and write /tests in Chat
  - Inline
    - Open the test class, write the test method name, let Copilot complete the body
  - Comment-driven
    - Write
    - // Test that findByEmail returns empty when user does not exist
  - Generates tests reflect the method signature and naming, not the actual logic
    - The generated tests must be reviewed to verify they test the right behaviour
    - Tests that pass without catching the real behaviour are worse than no tests



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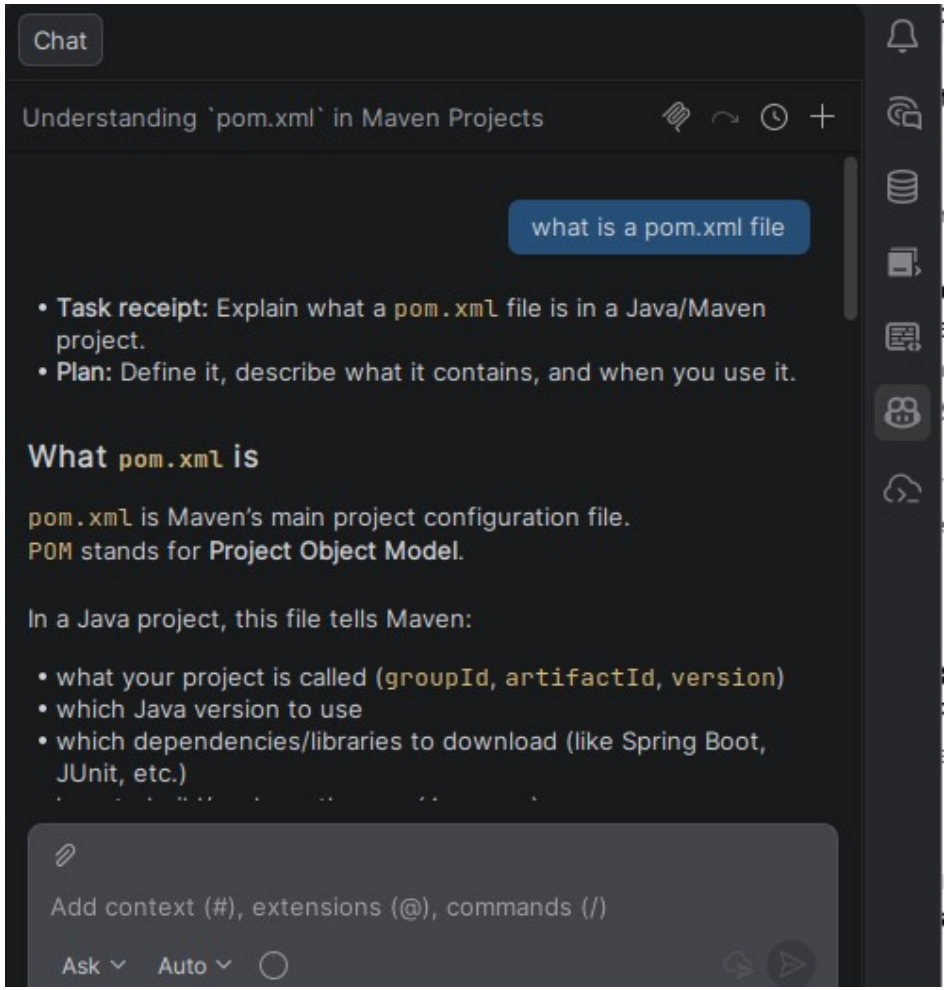
Writing Tests

- Copilot can generate a set of JUnit test methods
  - They compile, run, and pass
  - But that do not actually verify the contract of the method under test
  - Generates tests based on the method signature and name not on the implementation
  - A common pattern
    - Copilot generates an assertEquals that tests the exact value returned by the real implementation
    - Rather than testing what the value should be according to the specification
    - The result is a test that always passes



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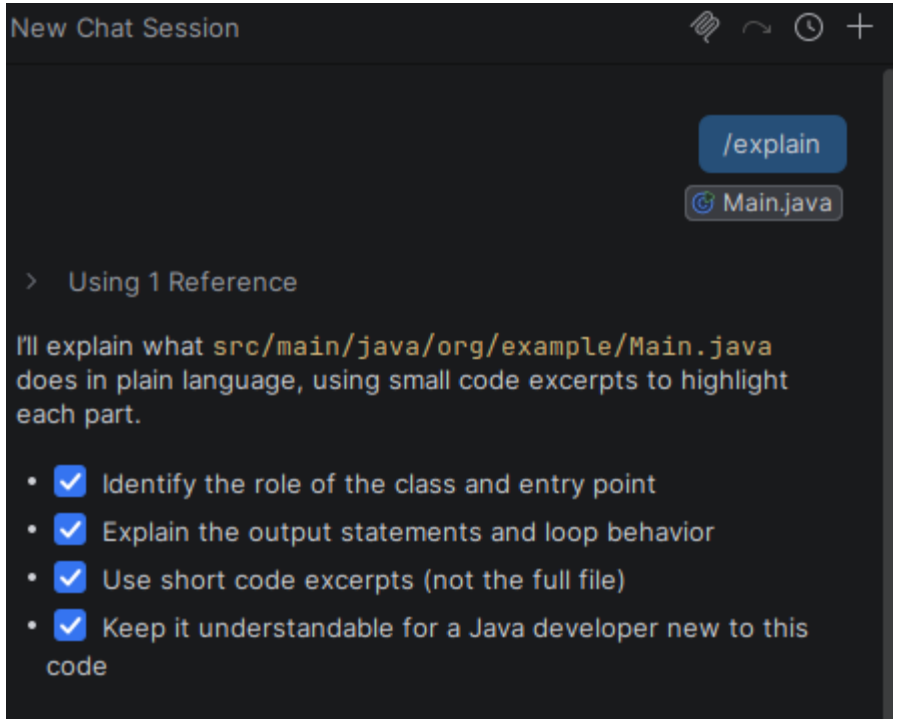
Explaining Unfamiliar Code

- Workflow
  - Select code you do not understand
    - Right-click → GitHub Copilot → Explain This
  - Or: type /explain in the Chat panel with code selected
- Useful for
  - Understanding a complex Spring Security filter chain
  - Decoding a multi-step Stream pipeline
  - Interpreting a JPA query with complex join conditions
  - Understanding the effect of an unfamiliar annotation



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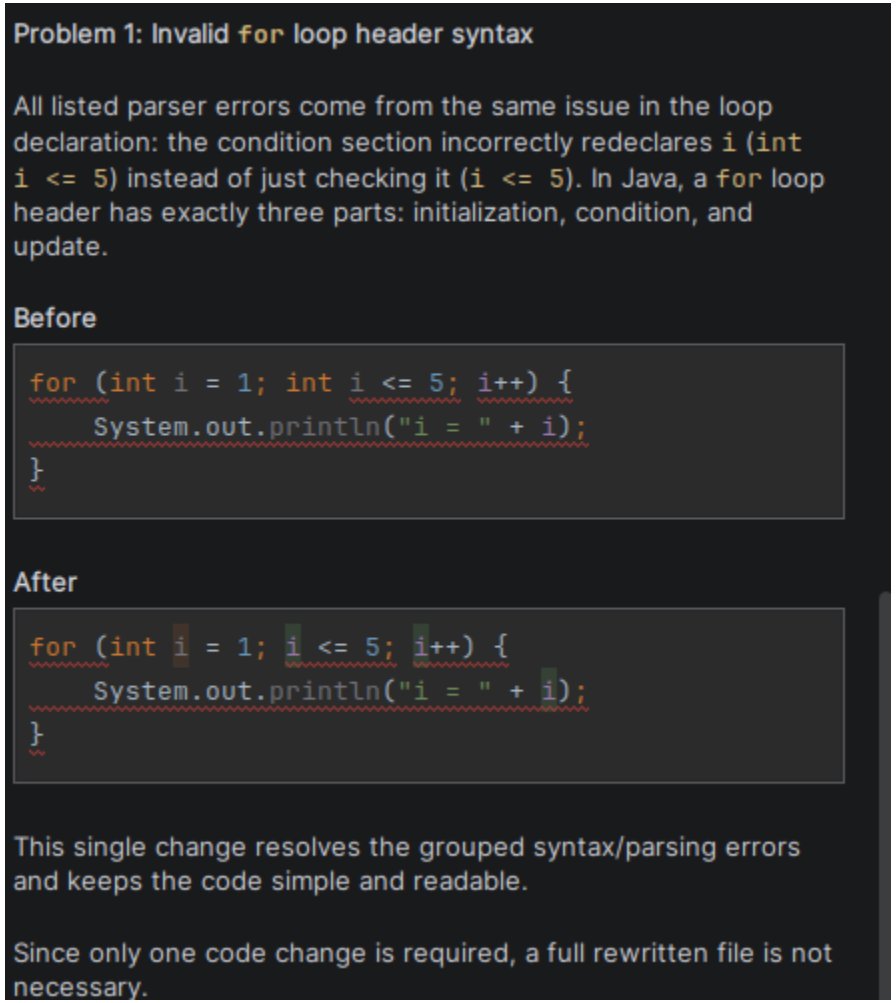
Fixing Errors and Refactoring

- Fix an error
  - Select
    - error code → right-click → GitHub Copilot → Fix This
    - /fix in Chat
    - Copilot proposes a correction and explains the change
    - Most useful for compilation errors and simple logic mistakes
- Refactoring via Chat
  - Select a method
    - Ask "Refactor this method to reduce its cyclomatic complexity"
    - Or: "Extract the validation logic into a separate method"
    - Or: "Rewrite this using Java Streams instead of a for loop"



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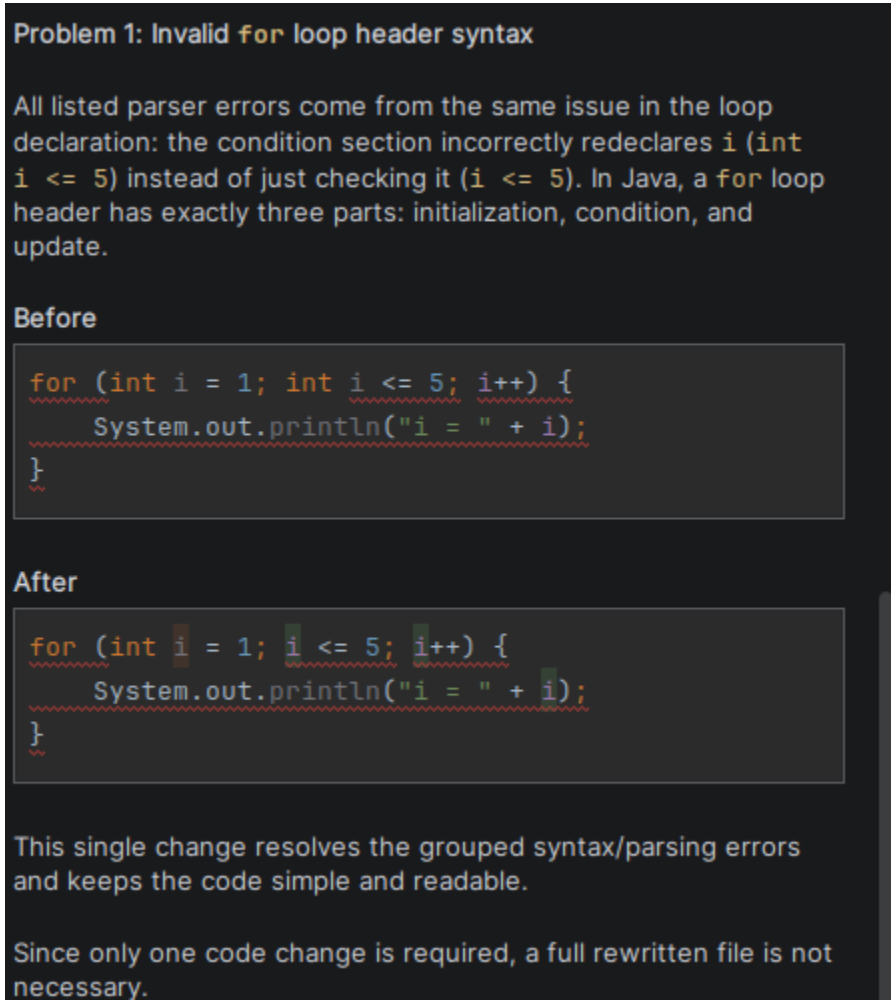
Fixing Errors and Refactoring

- Important:
  - Refactoring via Copilot does not replace IntelliJ's semantic refactoring tools
  - IntelliJ's Rename, Extract Method, etc. are guaranteed to compile
  - Copilot's refactoring suggestions must be compiled and tested before they are trusted



Problem 1: Invalid `for` loop header syntax

All listed parser errors come from the same issue in the loop declaration: the condition section incorrectly redeclares `i` (`int i <= 5`) instead of just checking it (`i <= 5`). In Java, a `for` loop header has exactly three parts: initialization, condition, and update.

Before

```
for (int i = 1; int i <= 5; i++) {
    System.out.println("i = " + i);
}
```

After

```
for (int i = 1; i <= 5; i++) {
    System.out.println("i = " + i);
}
```

This single change resolves the grouped syntax/parsing errors and keeps the code simple and readable.

Since only one code change is required, a full rewritten file is not necessary.

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Code Review and Validation

- Copilot Chat can be used as a first-pass review tool
  - Useful Chat prompts for validation:
    - "Are there any edge cases this method does not handle?"
    - "What happens if the input list is null or empty?"
    - "Is this implementation thread-safe?"
    - "Does this code have any potential resource leaks?"
    - "What are the security implications of this approach?"
  - Copilot's analysis is pattern-based and may miss issues that require runtime reasoning
- This is a supplement to, not a replacement for
  - IntelliJ's static inspections
  - Checkstyle rules
  - SAST tools (Checkmarx)
  - Human code review

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Code Review and Validation

- Valuable use of Copilot for validation
  - Asking targeted questions about specific concerns
    - "Are there edge cases this method doesn't handle?"
  - Is a better prompt than
    - "Review this code"
  - Because it directs attention to a specific class of issue
  - Copilot answers to these questions are starting points for human analysis
  - An answer of "this appears thread-safe" from Copilot is an invitation to verify that claim, not a confirmation

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Code Review and Validation

- What Copilot cannot validate
  - Copilot cannot verify correctness against requirements it has never seen
    - Business rules stored in external documentation
    - Non-functional requirements (performance thresholds, SLA constraints)
    - Compliance requirements not visible in the code
  - Copilot cannot detect all security vulnerabilities
    - It may generate vulnerable code while reviewing a prompt about security
    - It is not a substitute for dedicated SAST/DAST tooling (Checkmarx)
  - Copilot output is not subject to organizational policy or standards it has not been given
    - It does not know your team's Checkstyle rules unless you provide them
    - It does not know your organization's approved dependency list

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Code Review and Validation

- Quality gates
  - IntelliJ inspections, Checkstyle, JUnit test coverage, SAST scanning with Checkmarx
  - Are not replaceable by Copilot
  - They are deterministic, enforceable, and auditable
- Copilot
  - Is probabilistic, advisory, and not auditable in the same way
- Replacing formal quality gates with Copilot is any error
  - The correct mental model
    - Copilot accelerates the developer
    - The quality gate system ensures the output meets organizational standards regardless of how it was produced

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Code Review and Validation

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Prompt Engineering

Prompt Engineering

- Prompt engineering is the practice of designing inputs to an LLM to maximize output quality
  - it is closer to writing a precise specification or a clear brief by considering the prompt as a specification
  - Key insight: the model can only work with what it is given
    - Vague input → generic, often wrong output
    - Precise, contextual input → specific, often useful output
- Prompt engineering for code has two forms:
  - Inline context engineering
    - Shaping the code and comments the model can see
  - Chat prompt construction
    - Writing explicit, detailed prompts in the Chat panel

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Prompt Engineering

- Principle 1: Be specific about intent
  - State what you want, why you want it, and any relevant constraints
  - Vague prompt:
    - "Write a method to get users"
  - Specific prompt:
    - "Write a Spring Data JPA repository method that retrieves all active users whose email domain matches a given string, returns a paginated result, and sorts by last name ascending"
- Specificity applies to
  - The input type and constraints
  - The expected return type
  - Edge case handling (null input, empty result, invalid argument)
  - The framework or library to use

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Prompt Engineering

- Principle 2: Provide context explicitly
  - Do not assume Copilot can infer context it has not seen
  - Techniques for providing context
    - Open the relevant files before asking a Chat question (they enter the context window)
    - Reference specific files or symbols using # in Chat
    - Include relevant constraints in the prompt: "This service is called from multiple threads concurrently"
    - State the framework version: "Using Spring Boot 3.2 and Spring Security 6"
    - Paste the relevant interface or base class into Chat when asking about an implementation
  - The more relevant context the model has, the less it relies on generic patterns

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Prompt Engineering

- Principle 3: Use the code as context
  - The code in the current file is the most powerful context available
  - Name things deliberately
    - findActiveOrdersByCustomerId is better context than getOrders
    - CustomerValidationException is better context than AppException
  - Write the method signature before asking Copilot to complete the body
    - The return type, parameter types, and parameter names all signal intent
  - Write a Javadoc comment before the method
    - Copilot treats Javadoc as a specification to implement
  - Use @param and @return tags: they constrain output effectively

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Prompt Engineering

- Principle 4: Iterate, do not accept and move on
  - Treating the first suggestion as the final answer is the most common Copilot misuse pattern
  - Iteration strategies
    - Cycle through alternatives: Alt+] / Alt+[
    - Dismiss and retype with a more specific comment
    - Accept a partial suggestion, then ask Copilot to continue from the new position
    - In Chat: ask "Can you improve this to also handle the case where X?"
  - Iteration produces better results than re-prompting from scratch
    - The conversation history is part of the context
    - Follow-up prompts can refine rather than replace

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Prompt Engineering

- Principle 5: Validate all output
  - Every piece of Copilot output must be validated before it is committed to the codebase
- Validation checklist for generated code
  - Does it compile?
    - Copilot can generate code that references non-existent methods
  - Does it do what I asked?
    - Read the code, not just the structure
  - Are edge cases handled?
    - Null inputs, empty collections, boundary values
  - Does it follow project conventions?
    - Naming, error handling, logging patterns
  - Does it pass the tests?
    - Run the test suite; write tests if none exist
  - Would this survive a code review?
    - Apply the same standard you would to any committed code

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Prompt Engineering

- Principle 6: Avoid sensitive data in prompts
  - Copilot Chat prompts are sent to GitHub's servers
    - Do not include real credentials, connection strings, or secrets in prompts
    - Do not include personally identifiable information or sensitive business data
  - Practical rules
    - Use placeholder values in examples: "db_password_here", "customer-id-placeholder"
    - Ask conceptual questions rather than pasting data: "How do I handle a JWT with these claims?" not "Here is a real JWT token..."
    - Review your organization's Copilot usage policy for specific guidance on what data classifications are permitted
  - Code containing real infrastructure details (IP addresses, internal hostnames) should also be treated with caution

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Prompt Patterns

- Pattern 1: Specify the layer
  - "Write a Spring @Service method that..." vs. "Write a method that..."
  - The Spring annotation in the prompt activates Spring-specific patterns
- Pattern 2: State exception handling intent
  - "...and throw a ResourceNotFoundException if not found" produces correct exception handling
  - Without this, Copilot may return null or an empty Optional silently

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Prompt Patterns

- Pattern 3: Request tests alongside implementation
  - "Write the implementation and a corresponding JUnit 5 test class with Mockito"
  - Produces implementation and test in one step
- Pattern 4: Ask for explanations with code
  - "Write the implementation and add inline comments explaining each step"
  - Inline comments allow faster review and learning simultaneously

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Hallucination

- Hallucination in LLMs
  - Generating confident, plausible-sounding output that is factually incorrect
  - In code generation, hallucination appears as
    - Calling methods that do not exist on a class or library
    - Referencing API signatures that were changed in a different version
    - Generating import statements for packages that do not exist
    - Asserting incorrect behaviour for a framework or library
  - Detection
    - The compiler is the first line of defence, non-existent methods cause compilation errors
    - IntelliJ's inspections highlight unresolved references immediately
    - Code review against documentation catches API misuse
  - Response: reject the suggestion and re-prompt with the more specific API information

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Takeaways

- Copilot is a prediction engine, not a knowledge base or reasoning system
  - Quality of output is determined by quality of context
- Integration in IntelliJ provides two modes: inline completions and Chat
  - Both have distinct use cases and complement each other
- Prompts are specifications: precision and context determine output quality
  - Six prompt engineering principles
    - Be specific about intent
    - Provide context explicitly
    - Use the code itself as context
    - Iterate rather than accept and move on
    - Validate all output
    - Avoid sensitive data in prompts
  - Validation is non-negotiable
    - Copilot output must meet the same standard as human-written code

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Questions?